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The Taiwan Technology Industry Growth Premier Fund

Investment and Portfolio Management Final Presentation

112_2 Investment & Portfolio Management Group 04

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Agenda

- Investment Universe
- Investment Objective
- Return Back-testing
- Strategy Analysis
- Future Improvement
- Appendix

Our Mission

- **“Taiwan Technology Industry Growth Premier Fund”**
- **Taiwan**
  - **Participate in a market which outperforms the major stock markets in 2024.**
- **Technology**
  - **We participate in The key market sector that leads the economic trend in the world.**
- **Growth**
  - **We aim for finding the superior electronics tech firms that have growth potential.**
- **Premier**
  - **We enjoy better investing returns.**
  - **While we bear relatively low volatility.**

Key numbers for us to select the investment universe

35% / 12.36%

The contribution of manufacturing industry and IC industry to Taiwan's GDP. (2023)

30% / 3.43%

The return of Taiwan stock markets for the past one year and the projected real GDP growth of Taiwan in 2024.

59% / 39%

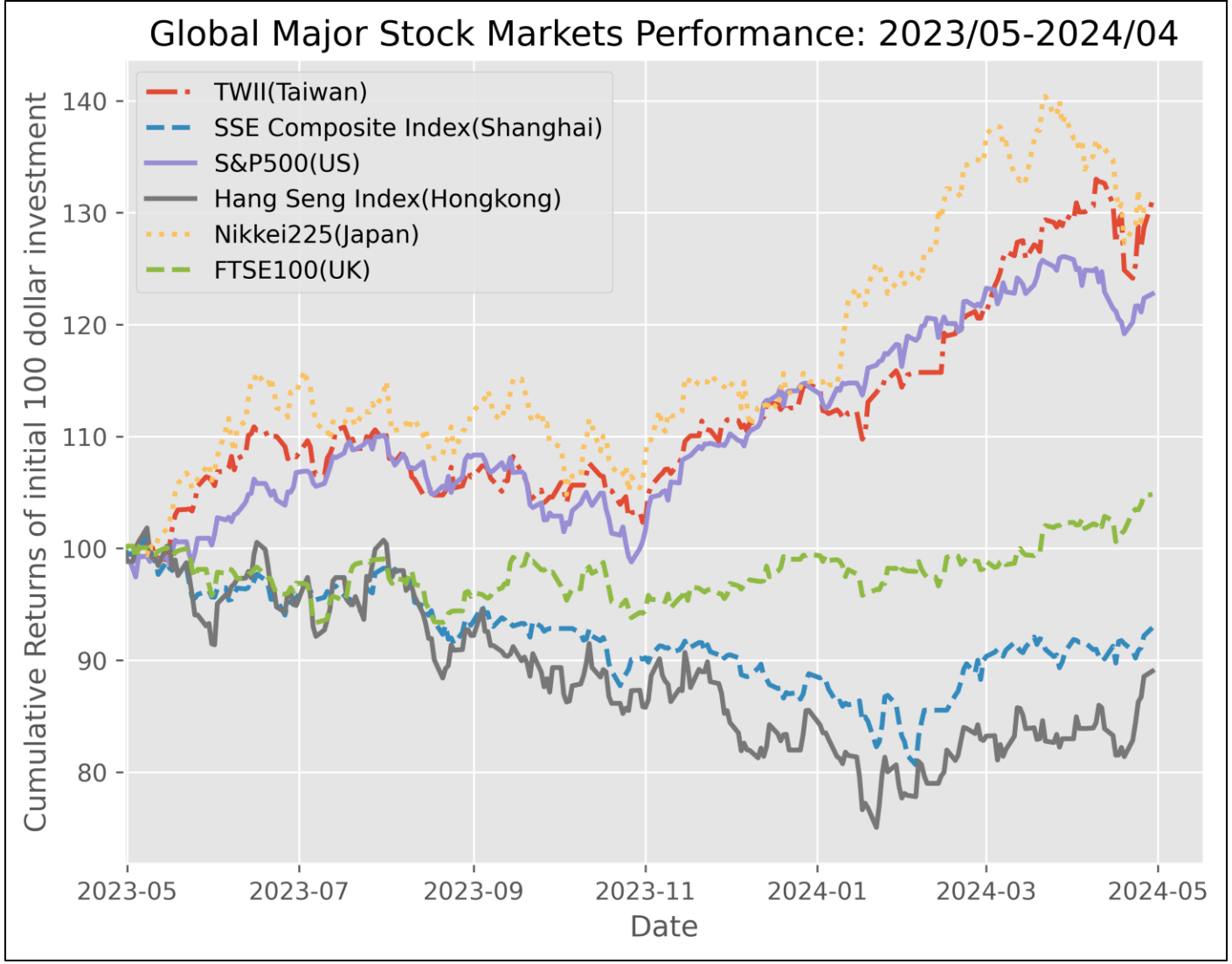
The percentage of electronics industry and semiconductor industry to the total market value in Taiwan stock exchanges. (2023/12)

Source: Taiwan Stock Exchanges, National Development Council (Taiwan), Ministry of Economics Affairs (Taiwan), World Economic Outlook Database April 2024

The Bloom of Taiwan stock markets and real economy for the past one year

- **Strong stock market:** Taiwan stock market had an astonishing return of about **30%** and outperformed US S&P500 index by **8%** for the past one year.
- **Healthy real economy:** In **2023Q4**, Taiwan's real gross domestic product (GDP) grew by **4.93%** from the same quarter of the previous year. Meanwhile, the real GDP is projected to grow by **3.43%** in 2024, which is **second highest** for the past 10 years in Taiwan. At the same time, the U.S. 2024 estimated real GDP growth is 2.4%.

Taiwan stock market outperforms other major markets

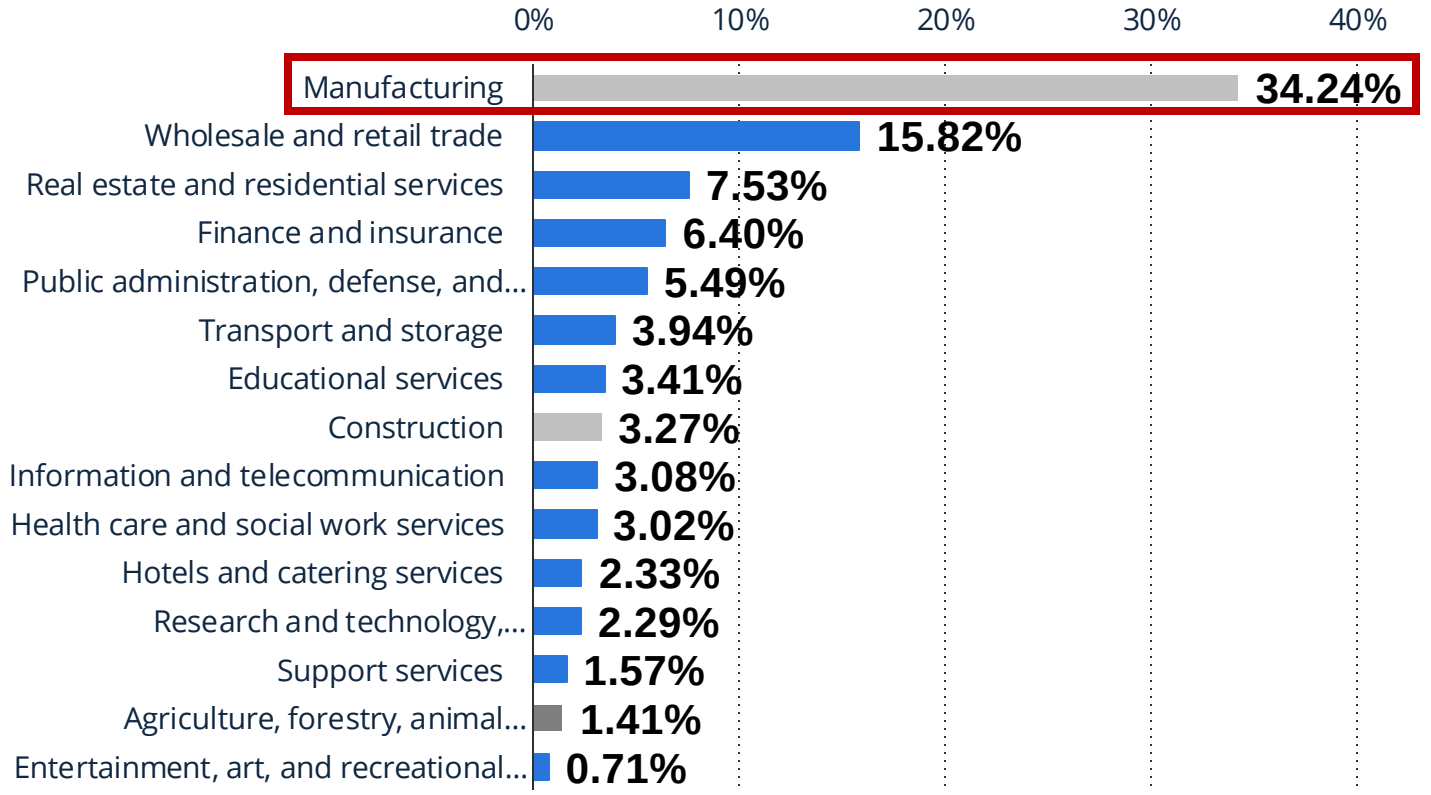


Source: Taiwan Stock Exchanges, Taiwan National Statistics, World Economic Outlook Database April 2024

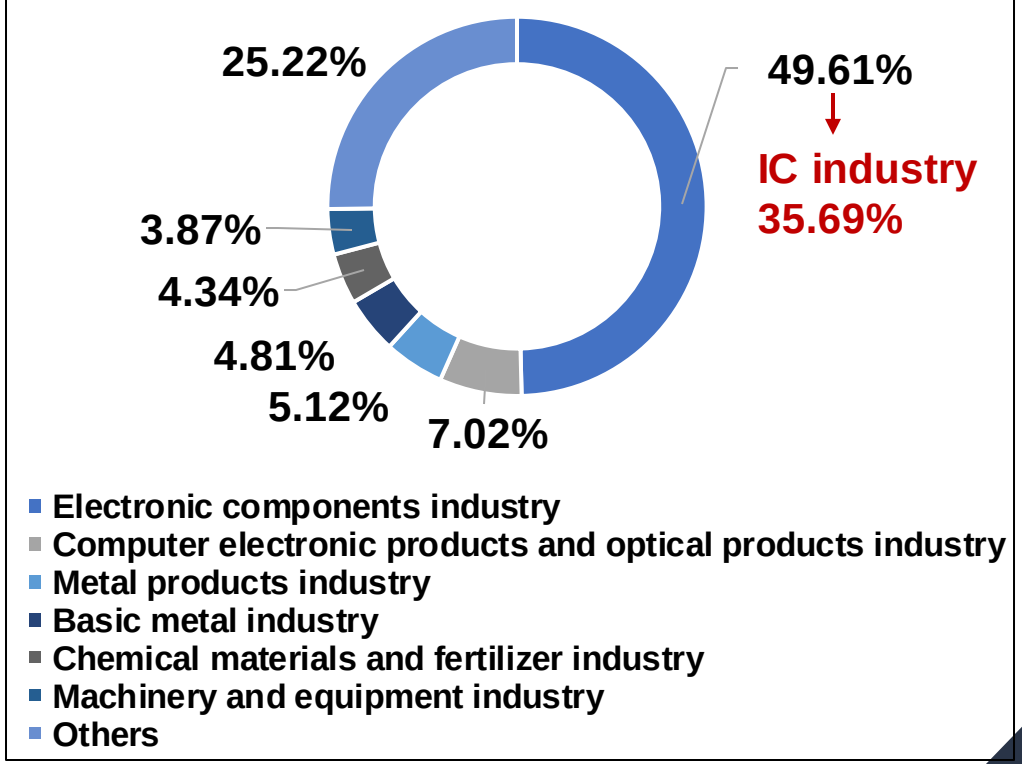
Manufacturing and Electronic components industry continue to lead Taiwan's economy

- The manufacturing industry accounted for about **35%** of the GDP in Taiwan in 2023, ranked 1st in all industry. Its performance has the biggest impact to the whole economy.
- The major components of manufacturing industry in Taiwan is **Electronic components (電子零組件) industry** which accounted for **49.61%**. Among them, the integrated circuit (**IC**, 積體電路) industry had a percentage of 71.95%. In other words, **the IC industry alone contributes about 12.36% of Taiwan GDP.**

GDP distribution in Taiwan in 2023, by industry



Taiwan Manufacturing by segments

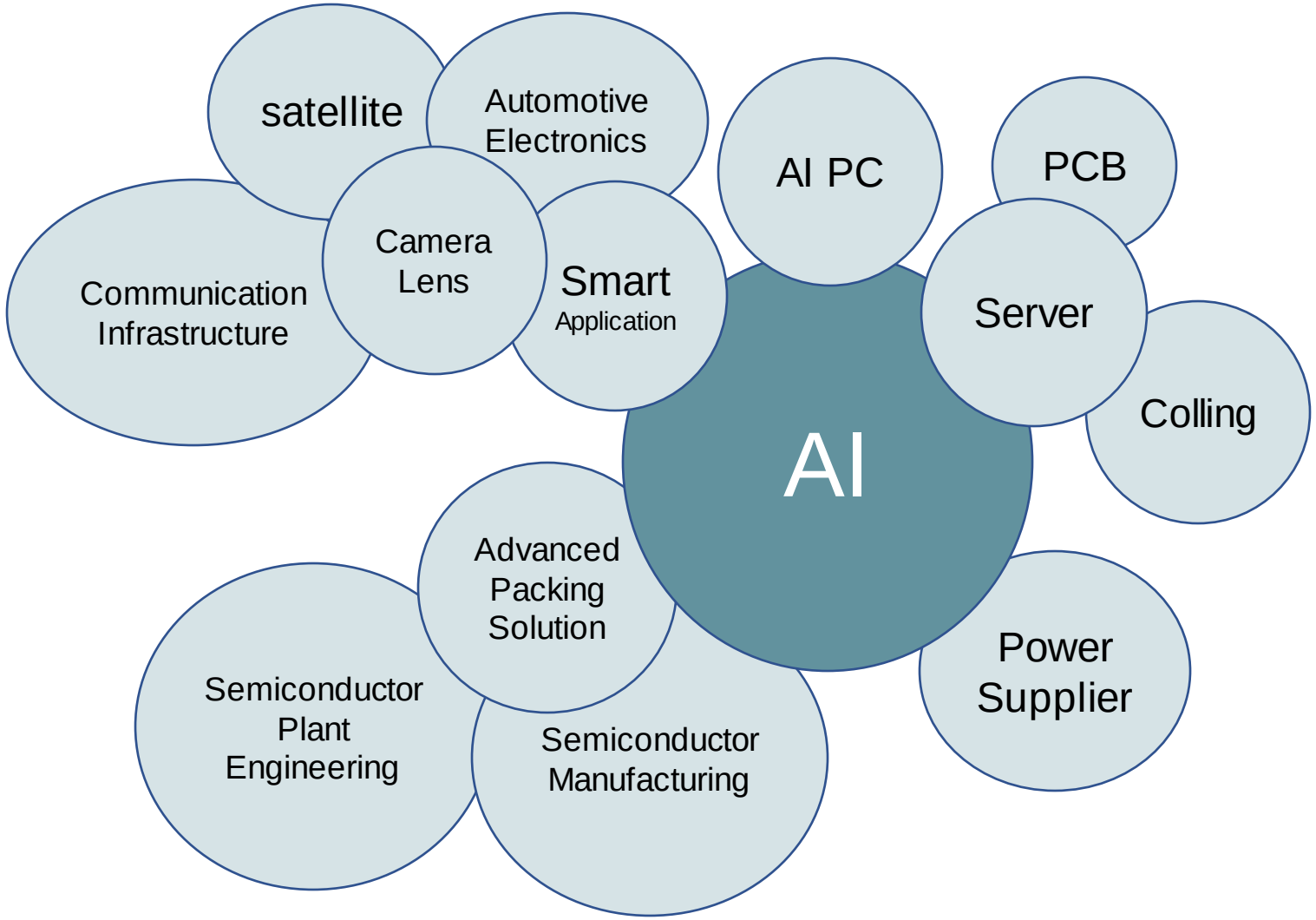


Source: National Development Council (Taiwan), Ministry of Economics Affairs (Taiwan)

We target our investment universe to the electronic technology industry

- **The dominance of electronic tech stocks in TWSE**
  - **59%:** electronic
  - **39%:** semiconductor
- **Intra industry diversification**
  - A wide range of application
- **Stock pitching criteria**
  - Stock **liquidity**
  - **Financial** performance
  - **Growth** potential
- **Further fundamental analysis**
  - Appendix

The major ending use of our stocks portfolio from TWSE

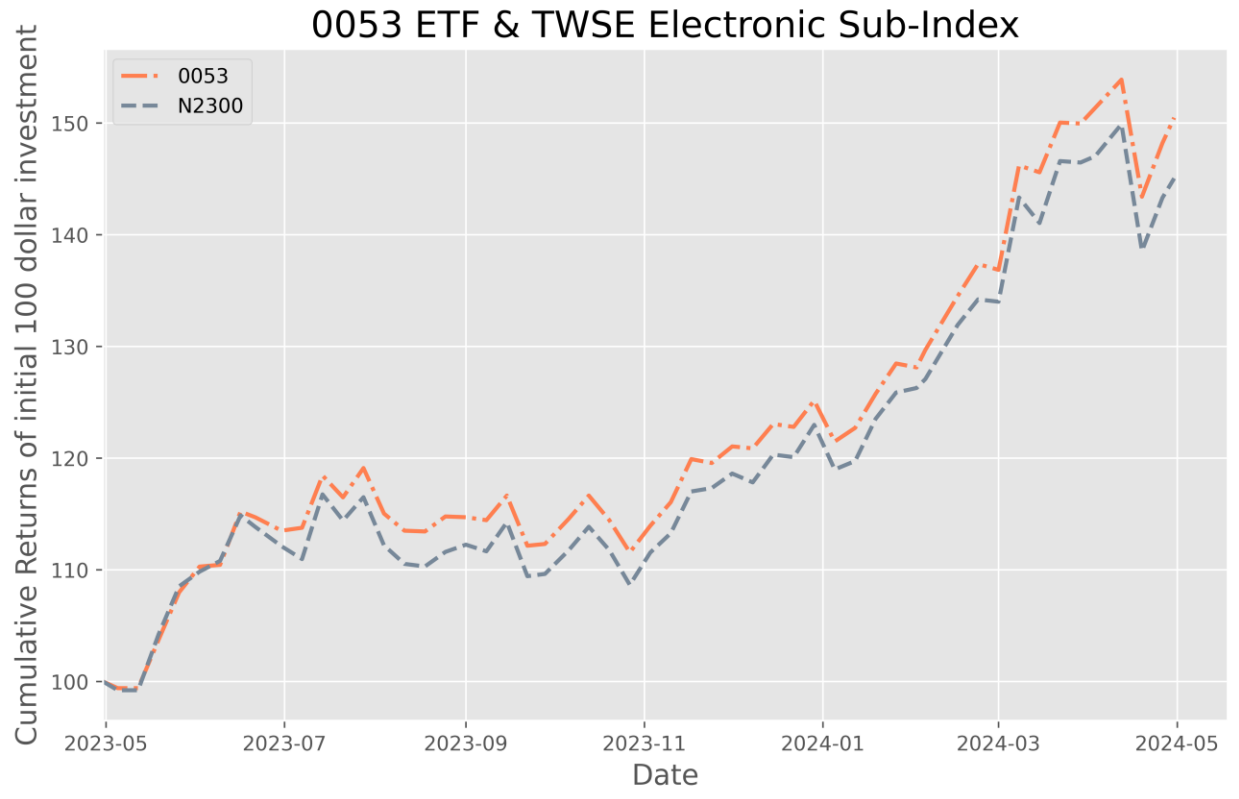


Source: TWSE

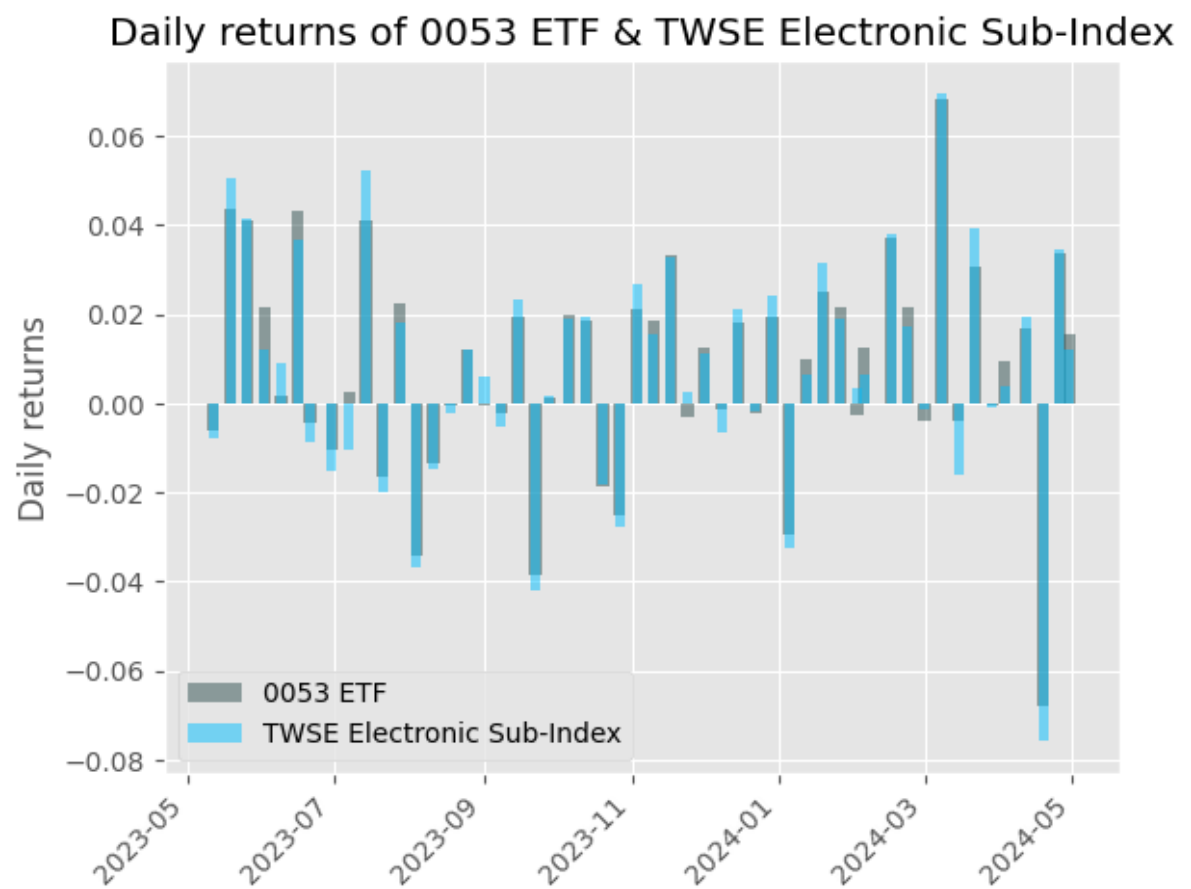
We will use Yuanta Taiwan Electronics Tech ETF as the benchmark to compete with

- Our benchmark index**:** Our fund aim to beat the benchmark index, **TWSE Electronics Sub-Index**, which measures the overall performance of Taiwan's electronics stocks.
- **Proxy of our benchmark index: Yuanta Taiwan Electronics Tech ETF (0053)** tracks the TWSE Electronics Sub-Index**,** so we will use this ETF as the benchmark to beat in our remaining presentation.

0053 ETF as the proxy of our benchmark index



The tracking error is neglectable



Source: TEJ, Yahoo Finance

Snapshot of Taiwan Technology Industry Growth Premier Fund

Span across

8

Sectors in Technology Industry

28%

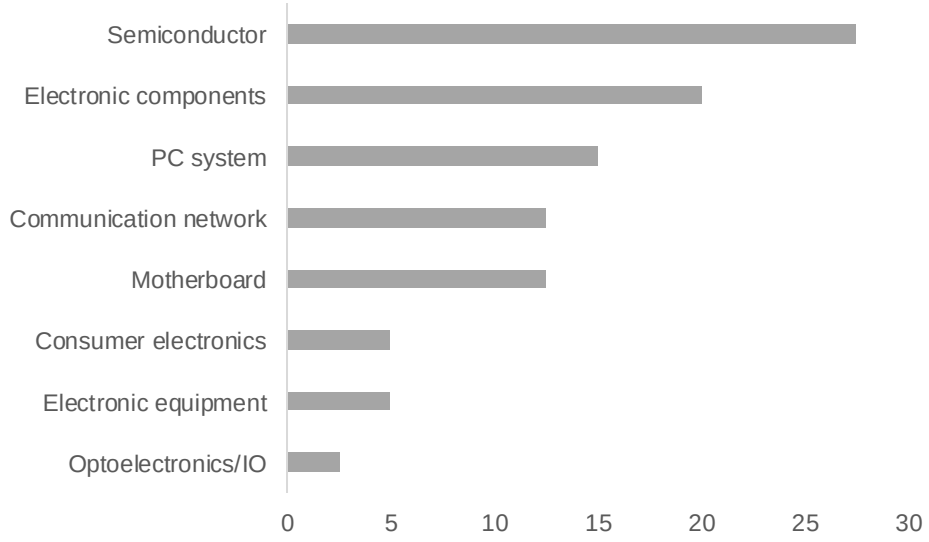
Semiconductor Sector

RR5

Risk Profile

40

Number of Holdings



Medium to Large

Market Capitalization

Aims to achieve long-term capital growth through investing in technology sector

Investment Policy Statement

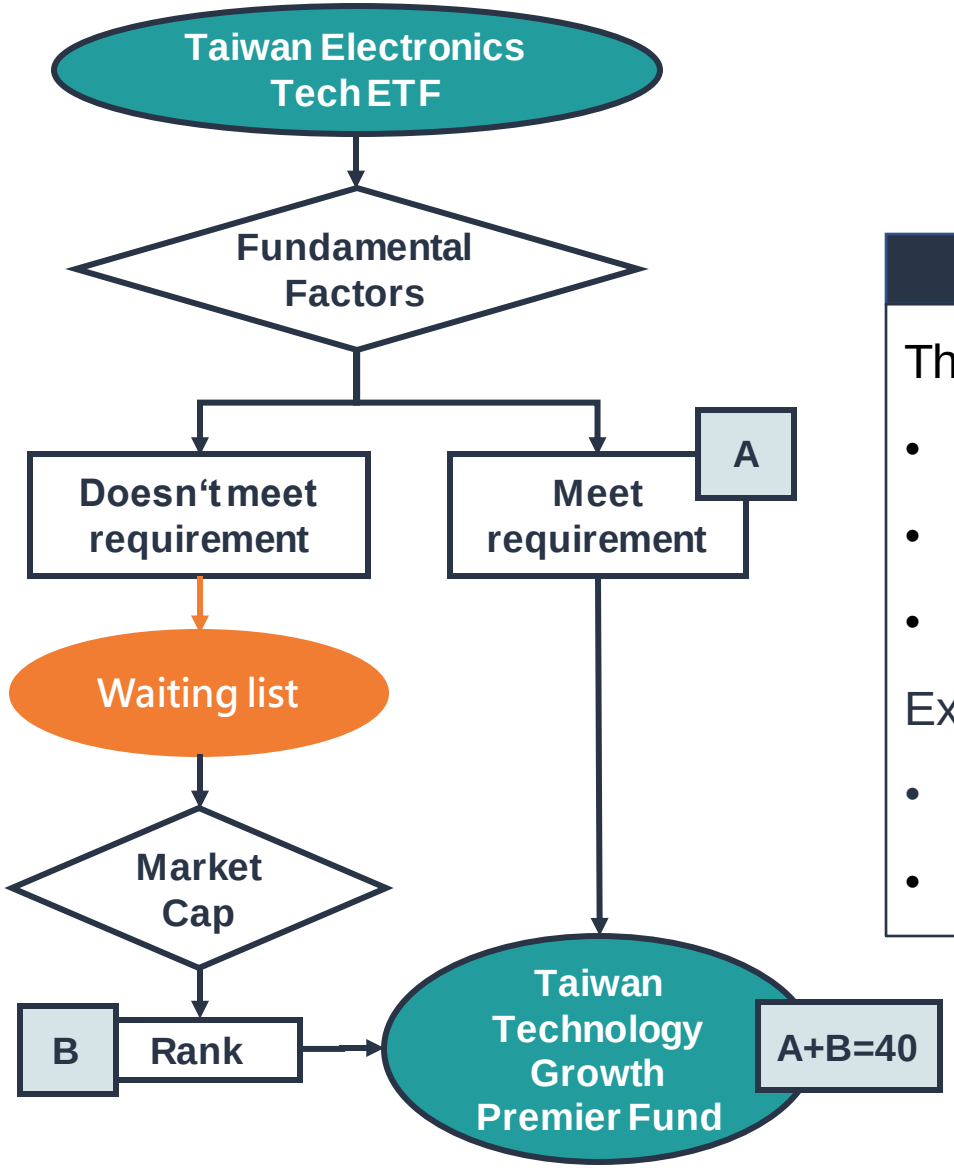
Investment Purpose

The investment philosophy of the Taiwan Technology Growth Premier Fund is driven by bottom-up stock selection within a **theme-based framework**. The Strategy aims to **achieve long-term capital appreciation** through investing in the equity securities of companies whose predominant economic activity is in the technology sector.

Investment Scope

The managers identify the **Yuanta/P-shares Taiwan Electronics Tech ETF** (0053 TW) as an appropriate scope for equity securities of Taiwanese companies expected to benefit from the development, advancement, and use of technology and communication services and equipment. In general, the investment managers looks for companies we believe display quality management, robust growth prospects, strong market positioning and good return on capital investment.

Holdings Selection Process



Holdings Selection

The fund selects companies with

- 3-year average **return on equity** of 15% and above
- 3-year average **gross margin** of 10% and above
- 3-year average **sales per share** growth of 10% and above

Expand investment pool for diversification and liquidity

- Largest market capitalization in the **Taiwan Electronics Tech ETF**
- Fund reaches 40 holdings

Aggressive risk profile

Investor Profile

The Fund may appeal to Investors looking to:

- capital appreciation by investing in equity securities
- a growth investment in the technology sector in Taiwan
- invest for the medium to long term

Risk Profile

Risk Return Level: RR5 (Aggressive)

Risk measurement metrics:

Value at risk, Maximum drawdown, Volatility

PORTFOLIO CHARACTERISTICS

Rebalance: Rebalanced at the first business day of each month

Benchmark: Yuanta/P-shares Taiwan Electronics Tech ETF (0053 TT)

Weighting Factor: Equally weighted

Expected Return: Expected annualized rate of return of 6%

Short position allowed: 0%

Risk Considerations: Principal risks to the Fund's investment strategy: Concentration risk, Counterparty risk, Market risk, Equity risk, Liquidity risk, Smaller and Midsize Companies risk, Sustainability risk

Key numbers of Our Investing Strategy

110% / 29%

Final cumulative return/Annualized return over the past three years.

1.31

Sharpe ratio, calculating by annualized return and annualized volatility over the past three years.

-7% / -19%

VaR5%/Max drawdown in past three years.

Verify if Equal-Weighted Strategy Outperforms Benchmark Indices and Other Weighting Strategies

Target Strategy

Equal-weighted (EW) portfolio 40 selected stocks from technology industry

Compare

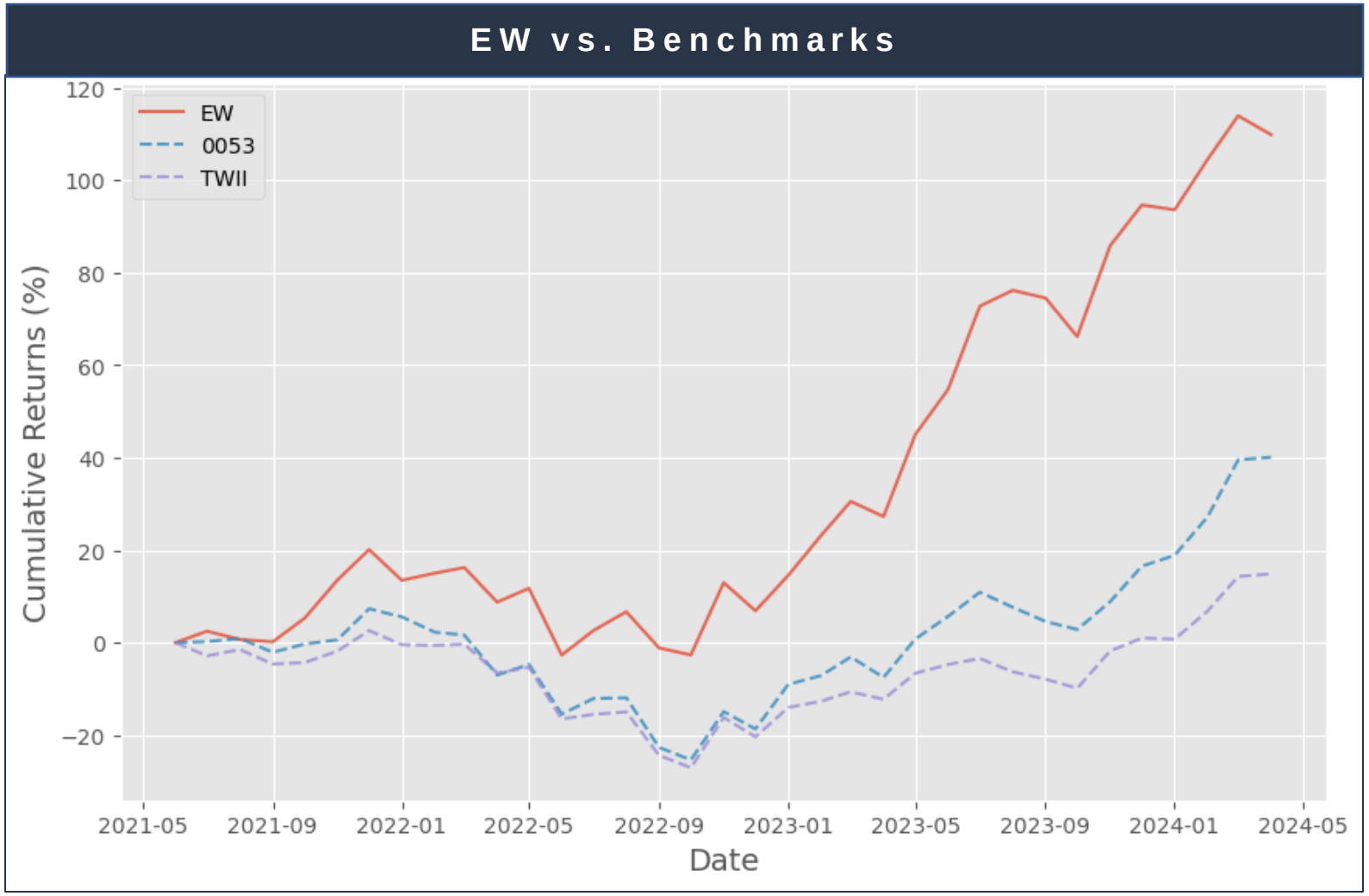
Benchmarks

- **Yuanta Taiwan Electronics Tech ETF (0053)**：Measures the overall performance of the Taiwan technology industry
- **Taiwan Weighted Index (TWII)**：Measures the overall performance of the Taiwan stock market

Other Weighting Strategies

- **Value-weighted (VW)**
- **Maximum Shape ratio portfolio (MSRP)**
- **Global minimum variance portfolio (GMVP)**
- **Long-only MSRP (MSRPnss)**
- **Long-only GMVP (GMVPnss)**

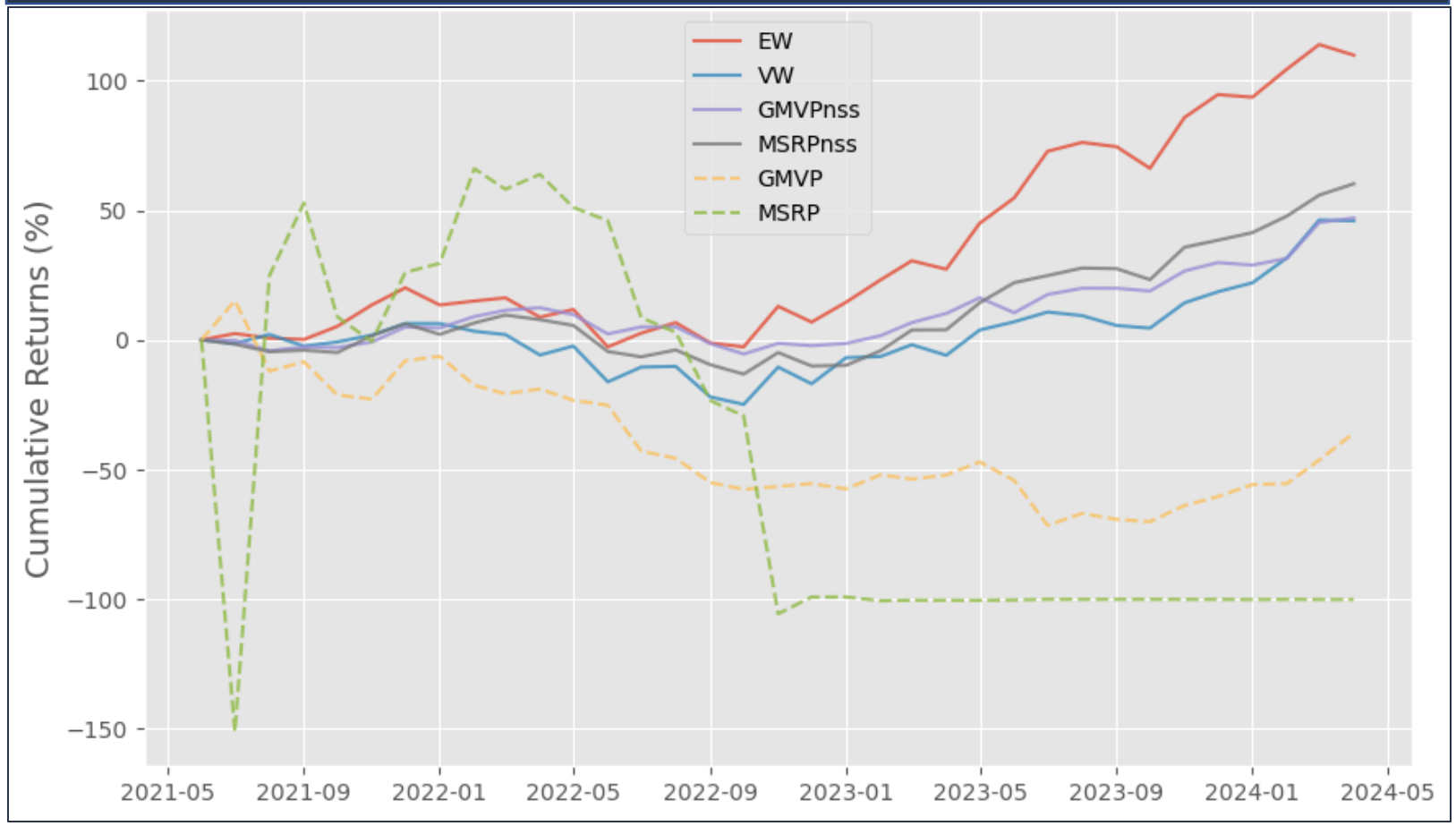
Equal-Weighted Strategy Outperformed the Benchmarks



- Equal-weighted portfolios outperformed benchmark 0053 and TWII since June 2021.

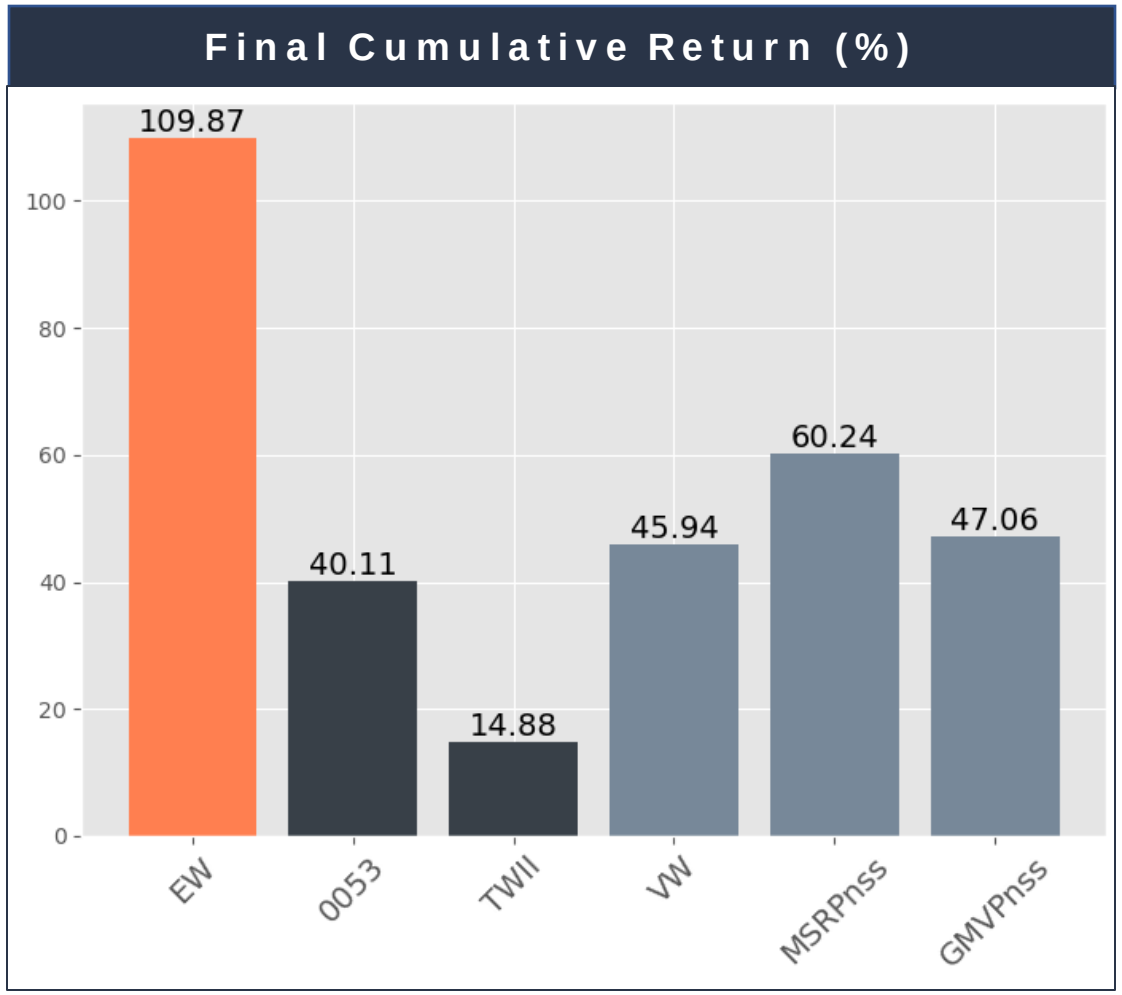
Equal-Weighted Strategy Outperformed Other Weighting Strategies

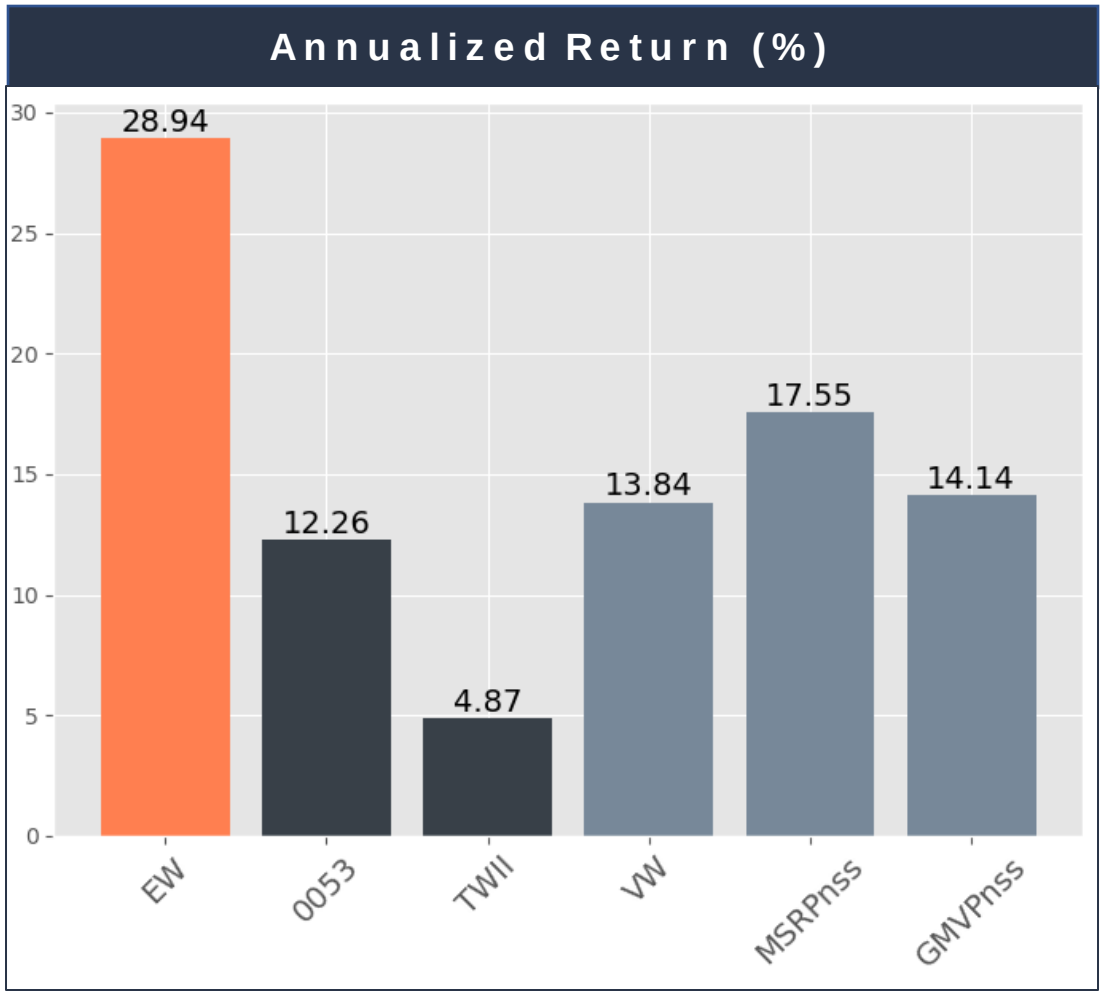
Weighting Strategy Comparison



- Equal-weighted portfolios outperformed other weighting strategies since August in 2022.
- MSRP, GMVP showed negative returns and high volatility in-sample period.

The Fund's Cumulative and Annualized Returns Outperformed the Benchmarks and Alternative Strategies

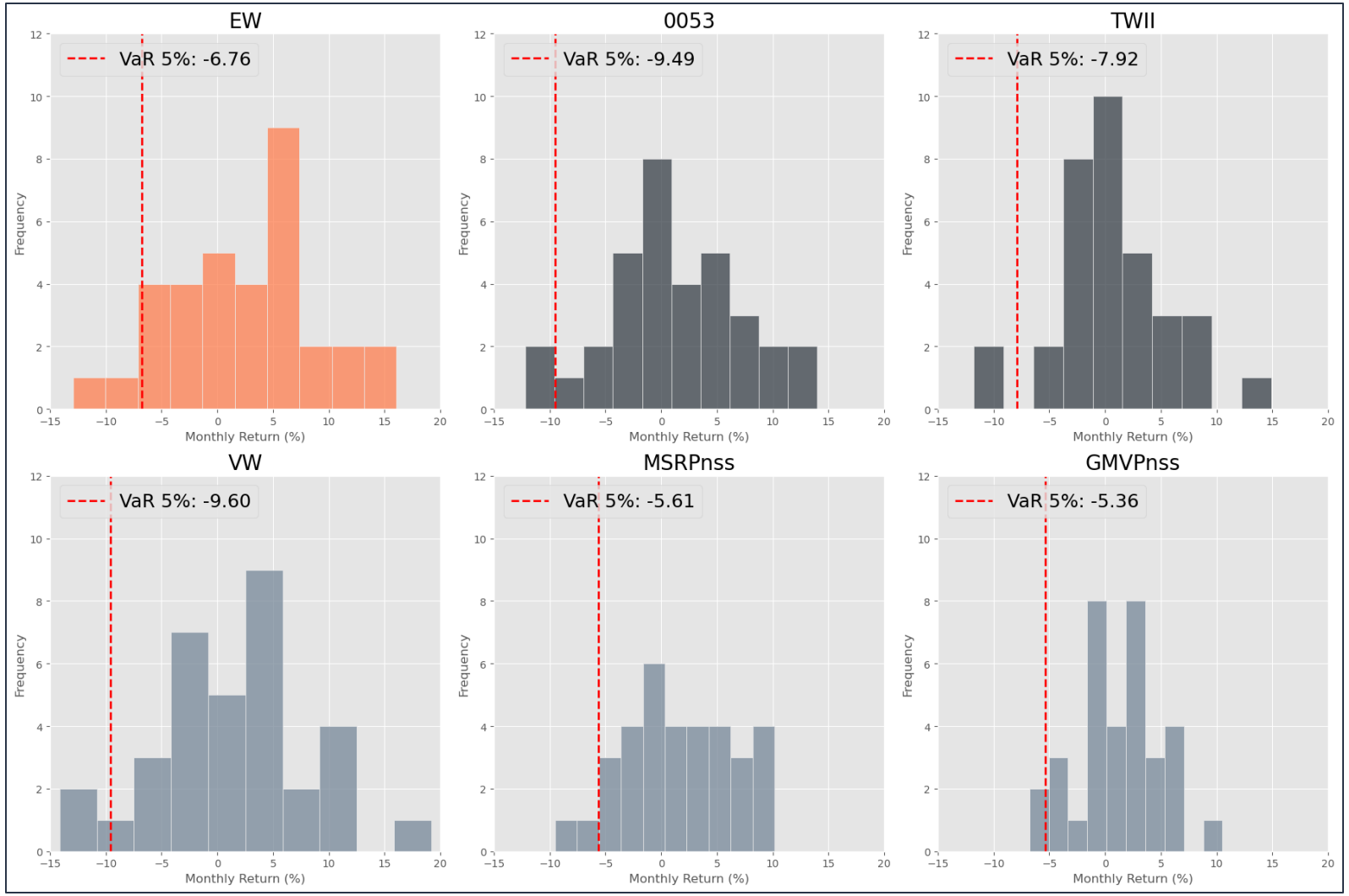




The final cumulative return of the fund over the past three years is **110%**, with an annualized return of **29%**, outperforming both the benchmarks and alternative strategies.

The Fund's 5% Value at Risk outperformed the Benchmarks

Monthly Return Distribution and VaR at 5% level



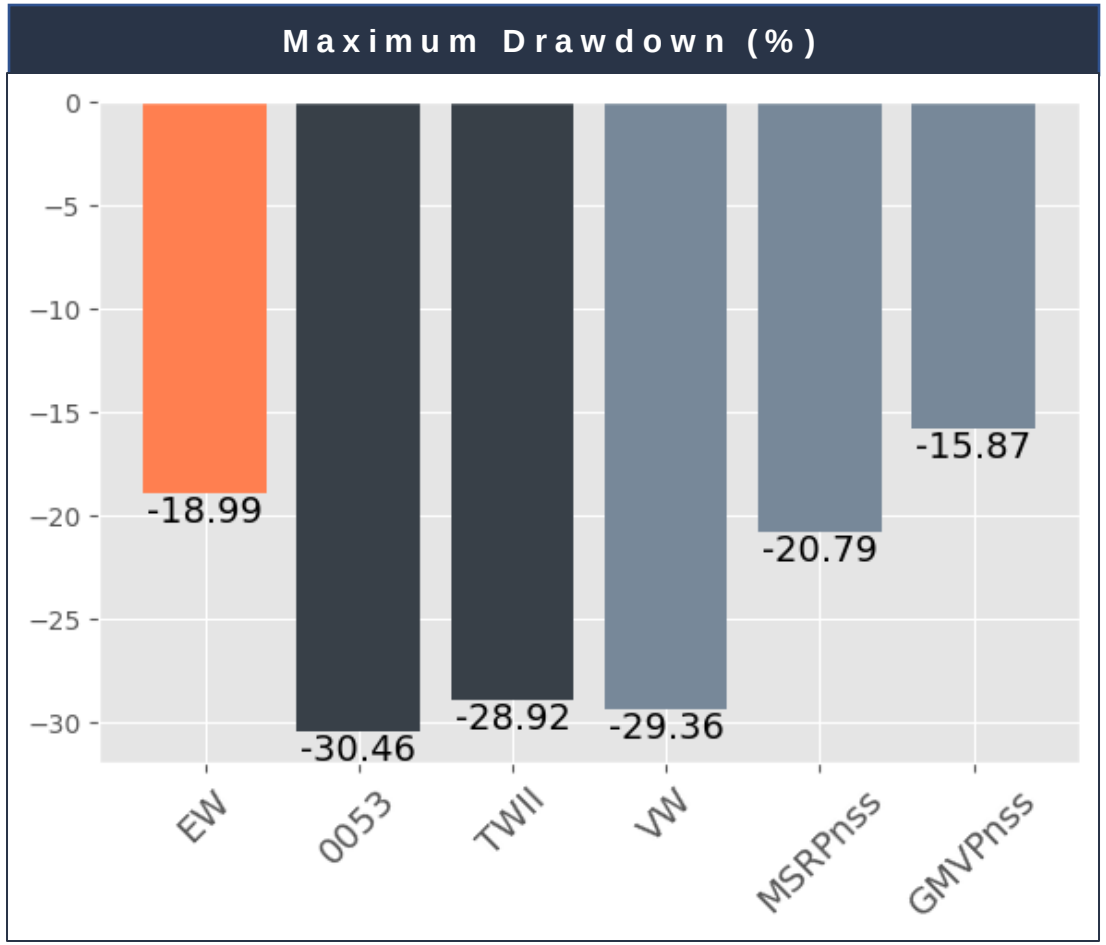
VaR and CVaR of the Fund

Level	VaR	CVaR
5%	-6.76%	-10.13%
10%	-5.49 %	-8.06 %
20%	-2.15 %	-6.42 %

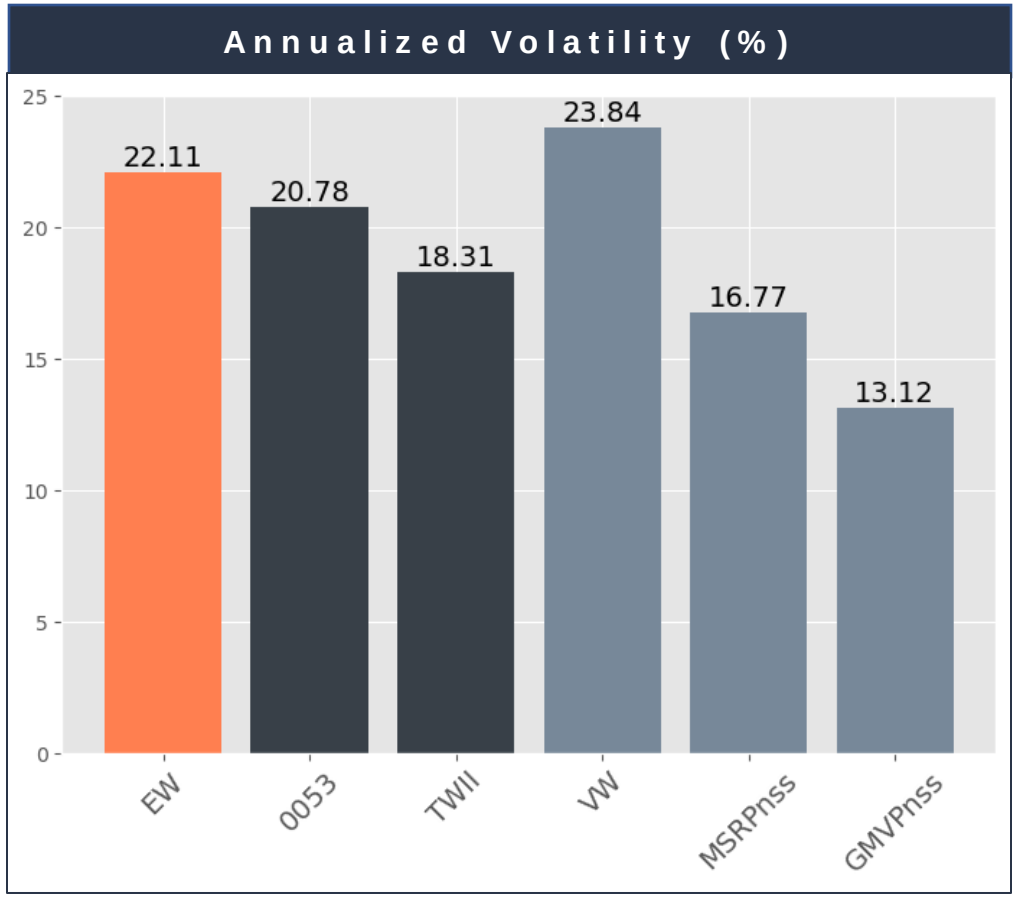
Over the past three years, the fund's 5% VaR is **-6.76%, outperforming the benchmarks and VW strategy** but underperforming the long-only MSRP and GMVP strategies.

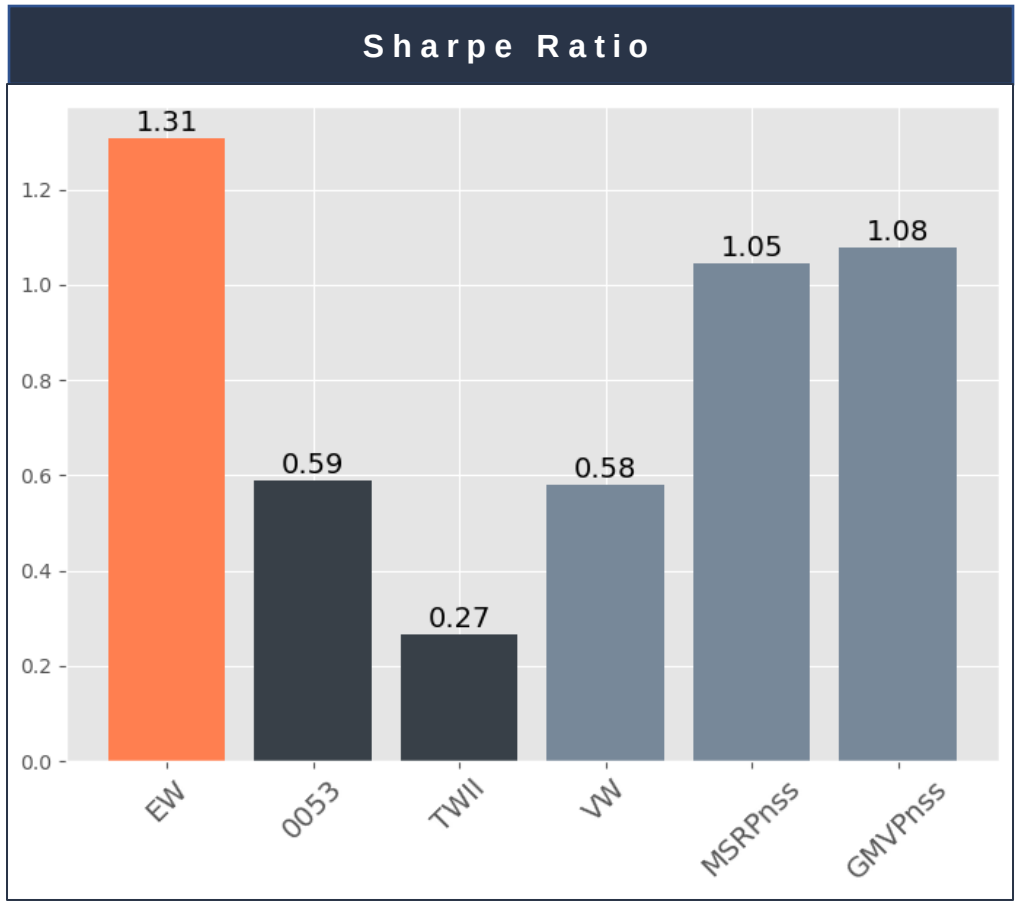
The Fund Exhibits Strong Performance in Terms of Percentage Maximum Drawdown

The fund's percentage maximum drawdown over the past three years is **-19%, outperforming the benchmarks**, and ranking **second among all strategies**, only slightly behind the long-only GMVP.



Despite Greater Volatility, the Fund Achieved Superior Risk-Adjusted Returns Compared to Benchmarks and Other Strategies



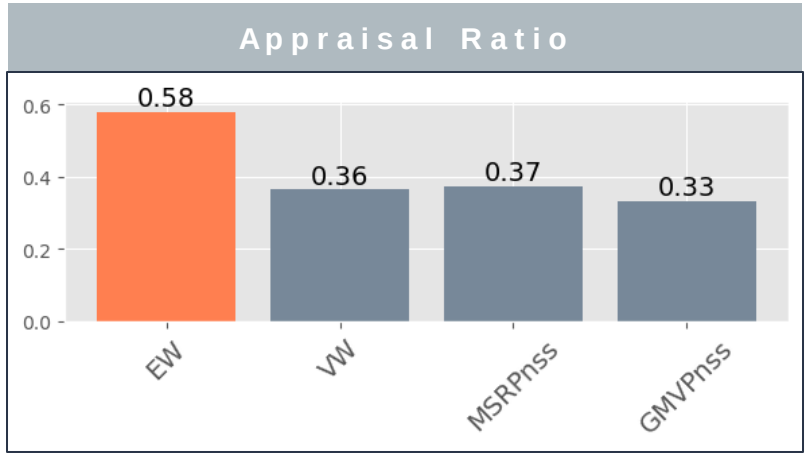


Despite having the **second-highest annualized volatility**, the fund achieved the **highest Sharpe ratio** compared to the benchmarks and other strategies over the past three years.

The Fund Outperformed the Market and Industry Benchmark with Significant Positive Abnormal Return and the Highest Appraisal Ratio Among all Strategies

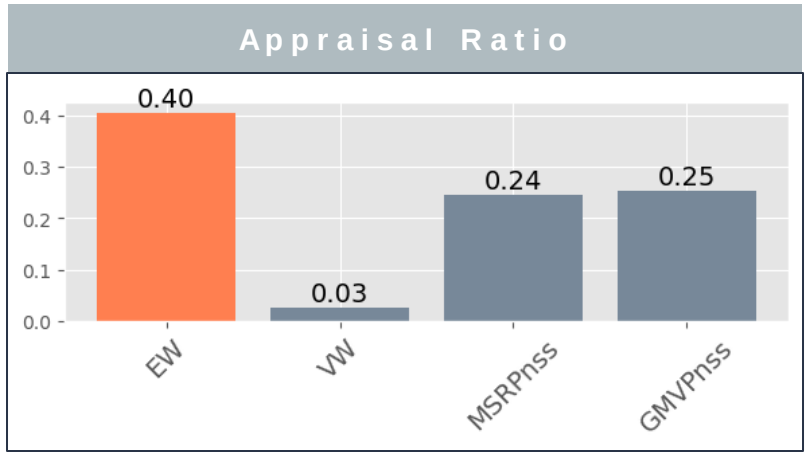
Market Benchmark TWII Regression Results

	Coefficient	Standard error	t-ratio	P>|t|
$\hat{\alpha}$	**0.018**	0.006	3.308	**0.002**
$\hat{\beta}$	1.050	0.106	9.944	0.000



Industry Benchmark Regression Results

	Coefficient	Standard error	t-ratio	P>|t|
$\hat{\alpha}$	**0.013**	0.006	2.276	**0.030**
$\hat{\beta}$	0.911	0.097	9.381	0.000



Using the TWII as the market return and 0053 as the technology industry-specific market return, our fund's **Jensen alpha** is **significant at the 1% and 5% levels**, respectively. The appraisal ratio in each case surpasses alternative strategies, highlighting our **superior active investment selection**.

Strategy Analysis Results from Backtesting: June 2021 to April 2024

Return Performance

Outperformed the benchmarks and all alternative strategies

- 110% Final Cumulative Return
- 29% Annualized Return

Risk Performance

Value at risk at 5% level and Maximum drawdown outperformed the benchmarks

- 22% Annualized Volatility
- -7% $VaR_{5\%}$, -10% $CVaR_{5\%}$
- -19% Maximum Drawdown

Reward to Risk Ratio

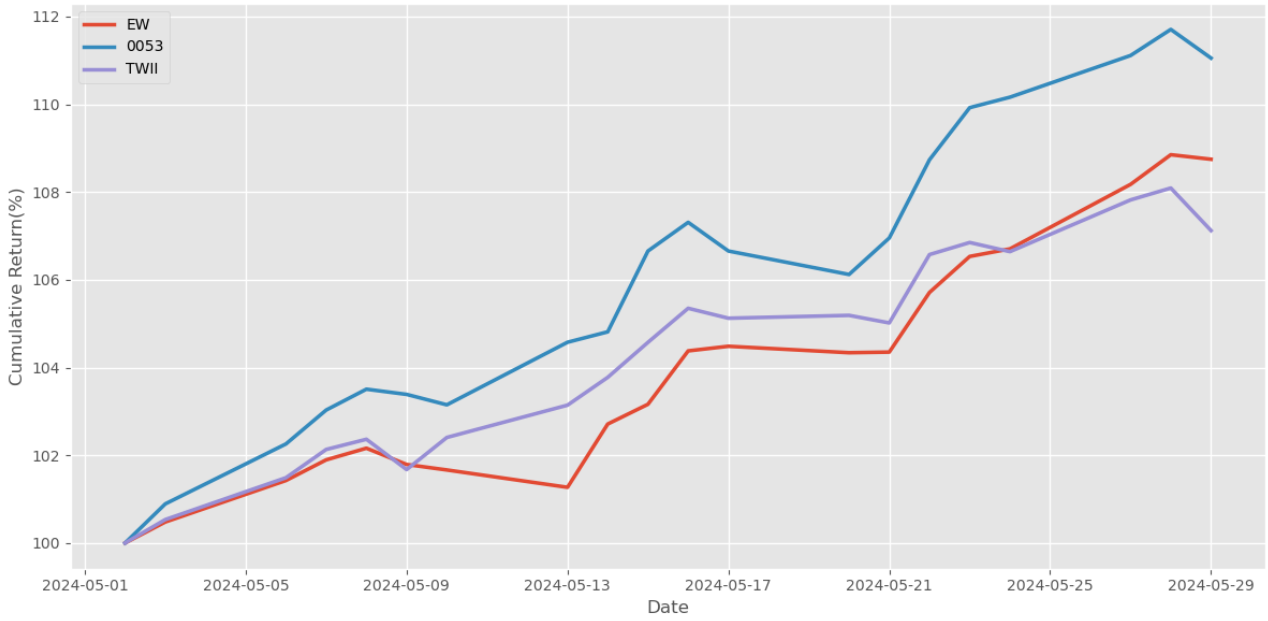
Beating both market and industry benchmarks with a significant Jenson alpha and superior appraisal ratio

- Sharpe Ratio: 1.31
- Jenson Alpha (market / industry): 0.02 / 0.01
- Appraisal Ratio (market / industry): 0.40 / 0.58

Our out of sample performance in May, 2024

Performance in May, 2024

- The Equal-Weighted Portfolio has shown a cumulative return about 8.5% in May.
- Although our portfolio did not outperform benchmarks 0050 and 0053 in May, it shows **potential** for long-term outperformance.
- Short-term performance consists of a relative large proportion of randomness. In contrast, this type of randomness have a average effect of zero in the long-run.



Take-away: A more complicated and challenging real world and introspection

Transaction cost

Marketing / Promotion

Seeking Alpha

"All models are wrong, but some are useful."

Source:

Take-away: A more complicated and challenging real world and introspection

- **Transaction cost**: without considering the transaction cost among rebalance processes, our
- **Products promotion and certain level of compromise**: In order to promote your fund product more smoothly, sometimes that you have to the preference of the majority of market investors.
  - e.g. promote product such as 00940 Yuanta Taiwan Value ETF(high dividend payments as Taiwan value).
  - Some popular product, for instance, factors ETF are popular in the U.S. Nonetheless, it is quite hard for us to find similar product in Taiwan even we believe that this is a good idea.
  - So what truly matters is what the market participants believe in rather than your personal opinion, even if you argue that you have a “right” point of view.
- **To beat the benchmark in the long run for a mutual fund is empirically difficult**
  - Our back-testing result
  - Our out-of-sample performance
- **“All model are wrong, but some are useful.”**
  - Impossibility of exist of a “perfect” model to describe the real world.
  - What we could do is to find a “useful” model to simplify our problem and provide insights.

Source:

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Source:

The complete list of the electronic stocks we picked

Stock	TWSE Ticker	TEJ Industry	Return-1Y (%)	3Y-average ROE (%)	3Y-average gross margin(%)	3Y-average sales growth per share(%)
台積電	2330	semiconductor	60.52	31.78	55.18	18.88
鴻海	2317	PC system	56.47	9.58	6.13	5.14
聯發科	2454	semiconductor	69.97	24.54	48.05	14.45
台達電	2308	Electronic components	9.88	16.74	28.90	12.63
廣達	2382	PC system	212.73	20.23	6.58	0.56
中華電	2412	communication network	1.54	9.61	36.40	2.44
聯電	2303	semiconductor	10.84	22.52	37.96	10.39
日月光投控	3711	semiconductor	58.19	19.47	18.41	7.98
緯穎	6669	PC system	111.20	35.31	8.56	12.56
聯詠	3034	semiconductor	61.43	48.89	45.99	16.99
台灣大	3045	communication network	5.00	17.55	19.45	11.43
國巨	2327	Electronic components	28.59	20.04	36.55	22.84
緯創	3231	PC system	150.27	16.49	6.99	1.43
華碩	2357	PC system	59.99	11.56	16.47	6.60
研華	2395	motherboard	15.29	23.94	38.75	8.62
智邦	2345	communication network	57.51	38.61	21.10	16.00
大立光	3008	consumer electronics	11.93	13.21	54.47	-4.03
創意	3443	semiconductor	38.90	41.73	33.23	26.54
光寶科	2301	Optoelectronics/IO	40.98	18.25	19.90	-1.45
欣興	3037	motherboard	31.23	25.07	26.01	9.13

Source: TEJ, Statementdog.com

The complete list of the electronic stocks we picked

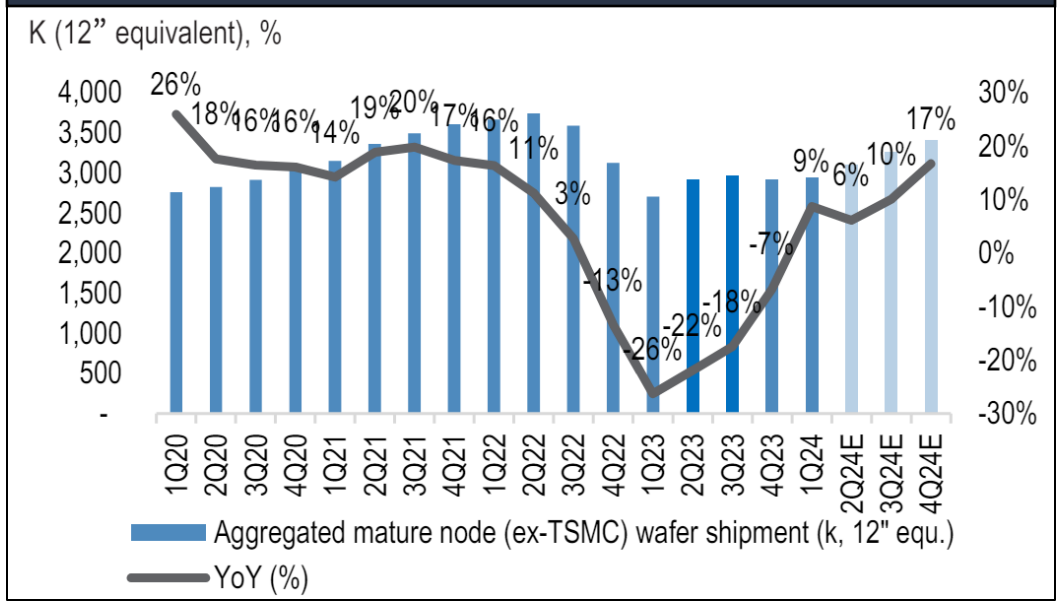
Stock	TWSE Ticker	TEJ Industry	Return-1Y (%)	3Y-average ROE (%)	3Y-average gross margin(%)	3Y-average sales growth per share(%)
和碩	4938	PC system	47.31	10.35	3.90	-3.34
瑞昱	2379	semiconductor	53.91	35.88	47.36	8.93
奇鋐	3017	Electronic components	321.10	27.48	19.34	14.45
技嘉	2376	motherboard	132.87	23.94	17.29	19.86
遠傳	4904	communication network	7.14	14.40	30.24	5.63
嘉澤	3533	Electronic components	69.72	24.33	43.65	13.58
南亞科	2408	semiconductor	0.48	5.99	21.91	-13.55
台光電	2383	motherboard	156.75	25.61	26.23	16.26
金像電	2368	motherboard	99.80	28.83	25.42	9.53
信邦	3023	Electronic components	-15.10	23.64	25.35	14.68
健策	3653	Electronic components	118.86	21.54	33.60	21.85
旭隼	6409	Electronic components	-9.85	49.73	29.44	13.87
智原	3035	semiconductor	84.99	21.24	47.91	33.43
漢唐	2404	Electronic equipment	95.67	36.89	13.61	34.20
玉晶光	3406	consumer electronics	22.68	15.58	36.98	10.96
新唐	4919	semiconductor	2.24	20.44	41.01	28.67
勤誠	8210	Electronic components	255.36	19.91	20.88	14.49
天鈺	4961	semiconductor	9.54	27.39	38.46	26.33
帆宣	6196	Electronic equipment	27.54	22.16	10.17	31.70
中磊	5388	communication network	56.61	17.57	14.22	21.88

Source: TEJ, Statementdog.com

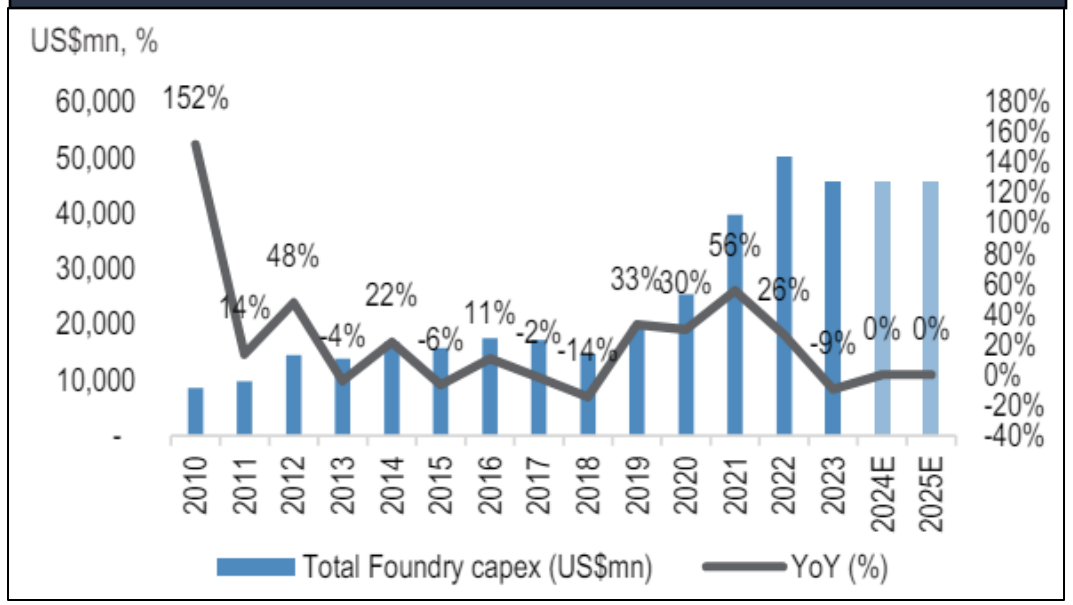
Signals to the bottom of semiconductor industry; slow recovery for AI / non AI demand

- **Demand bottomed out in 1Q24; continued recovery into 2H24:** Most foundries pointed to **a cyclical recovery into 2H24**, due to cautious inventory management from customers and slow end demand recovery.
- **Leading-edge demand to benefit from proliferating AI adoption:** We expect a smooth ramp for N3 in 2024 (Apple iPhone, iPad and Mac, Bitmain, QCOM, Intel), with incremental demand coming from new customers into 2025 (AI accelerators, NVDA, AMD, Mediatek, Broadcom). We expect N5 utilization to remain high throughout 2024 due to strong demand from AI accelerators (NVDA GPU, AMD GPU, AWS Inferentia/Trainium, Meta MTIA, MSFT Maia etc.) and higher adoption for ARM server CPUs (Amazon Graviton, Google Avion, MSFT Cobalt).

Blended mature Foundries' wafer shipment



Major Foundries' capex and YoY trend



Stocks list:

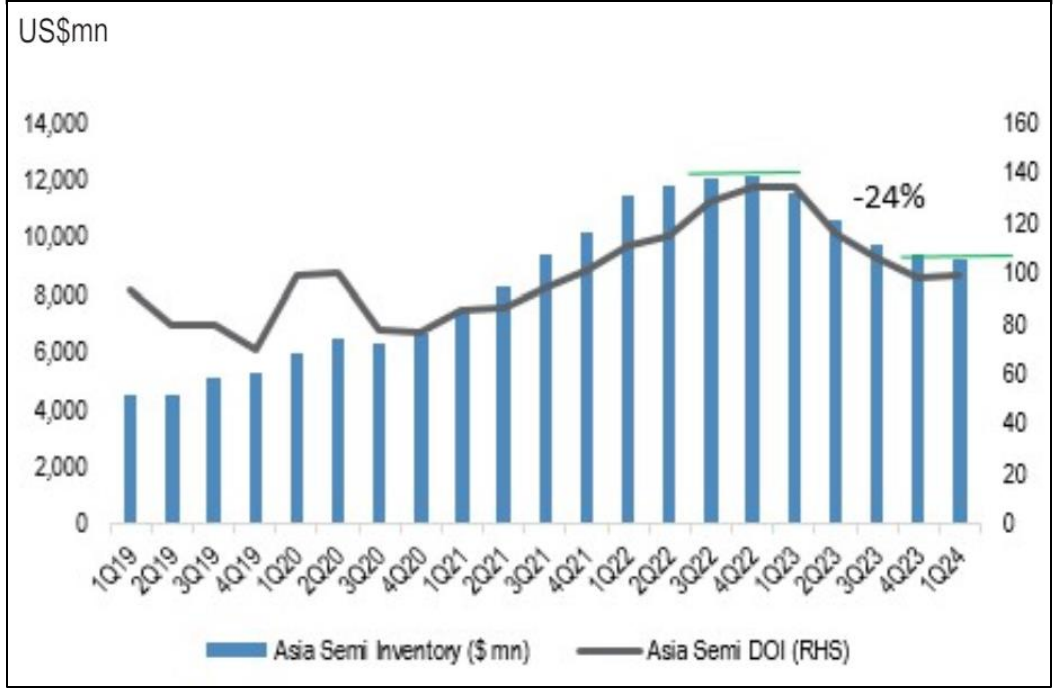
- ✓ 2330 台積電
- ✓ 2454 聯發科
- ✓ 2303 聯電
- ✓ 3711 日月光
- ✓ 3034 聯詠
- ✓ 3443 創意
- ✓ 2379 瑞昱
- ✓ 2408 南亞科
- ✓ 3035 智原
- ✓ 4919 新唐
- ✓ 4961 天鈺

Source: JP Morgan

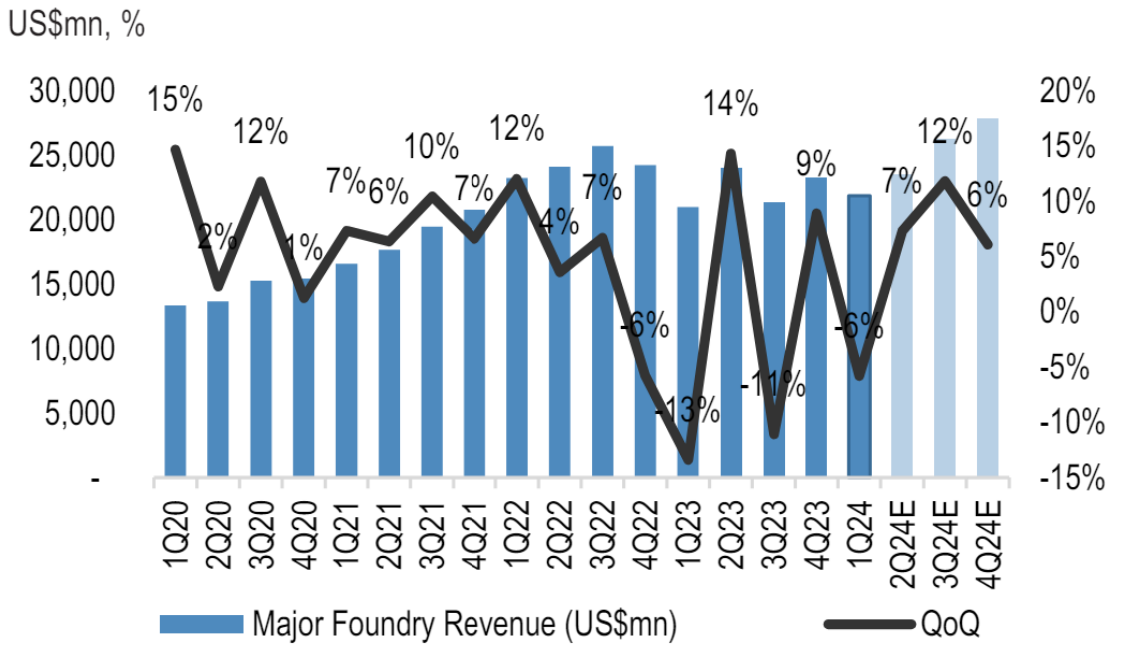
Signals to the bottom of semiconductor industry; slow recovery for AI / non AI demand

- **Inventory levels normalizing among Asia Fabless, but needs more time for international Fabless/IDM:** After six quarters of aggressive destocking since 2H22, inventory levels for Asian Semi vendors are down 24% from the peak with **DOI** also approaching normalized levels (99 days vs peak level of 134 days). At a global level, inventories continue to be high and may require another 1-2 quarters to reach normalized levels. **Rush orders** are emerging in selected areas, but overall, we expect restocking demand to broaden out in 2H24 for Foundries.
- Overall, we **prefer Asia pacific semiconductor foundries and fabless firms over U.S. firms**, so we set our investment universe in Taiwan stock market.

Asian Semi inventory and DOI



Major Foundries' revenue and QoQ trend



Stocks list:
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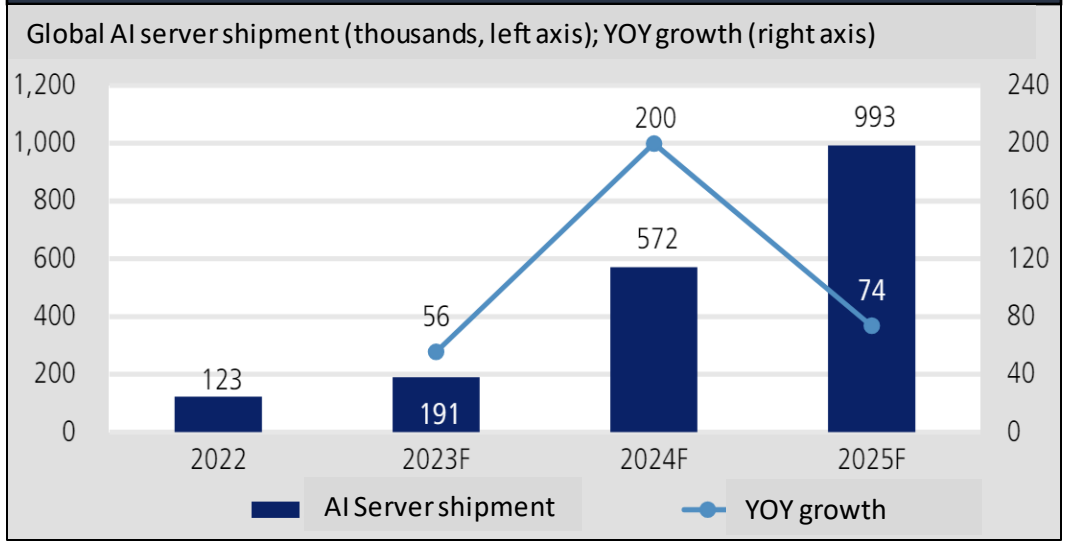
- ✓ 3711 日月光
- ✓ 3034 聯詠
- ✓ 3443 創意
- ✓ 2379 瑞昱
- ✓ 2408 南亞科
- ✓ 3035 智原
- ✓ 4919 新唐
- ✓ 4961 天鈺

Source: JP Morgan

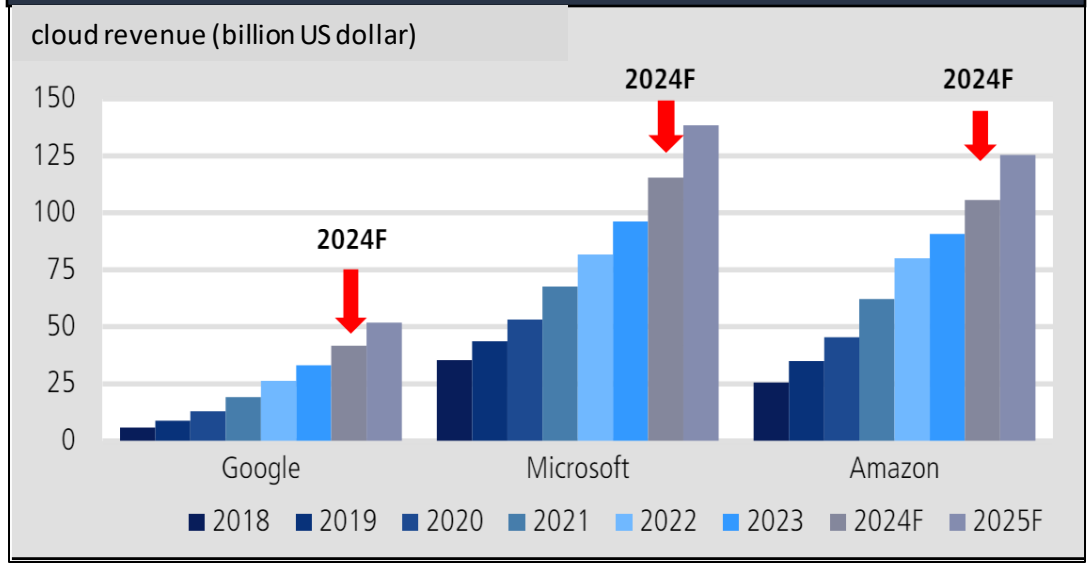
5G, AI, edge computing are keys ending applications in communication industry

- The 2024 Mobile World Congress (MWC) will be held from February 26 to February 29 held in Barcelona, Spain.
- This MWC exhibition focuses on 5G applications. Sercomm (5388) exhibited millimeter wave/Sub-6GHz Small base stations and FWA CPE; Accton (2345) focuses on mainly switch.
- Taiwanese electronics OEMs including Quanta (2382), Wiwynn (6669), and Gigabyte (2376) are all participating in the exhibition, and they continue to deploy **5G, AI, edge computing and liquid cooling technologies**. The products displayed hope that new technologies such as 5G and AI can bring changes to the manufacturing industry. In the future, AI can improve human work efficiency and making AI more human.

Global AI servers shipment



Major cloud service providers cloud revenue



Stocks list:

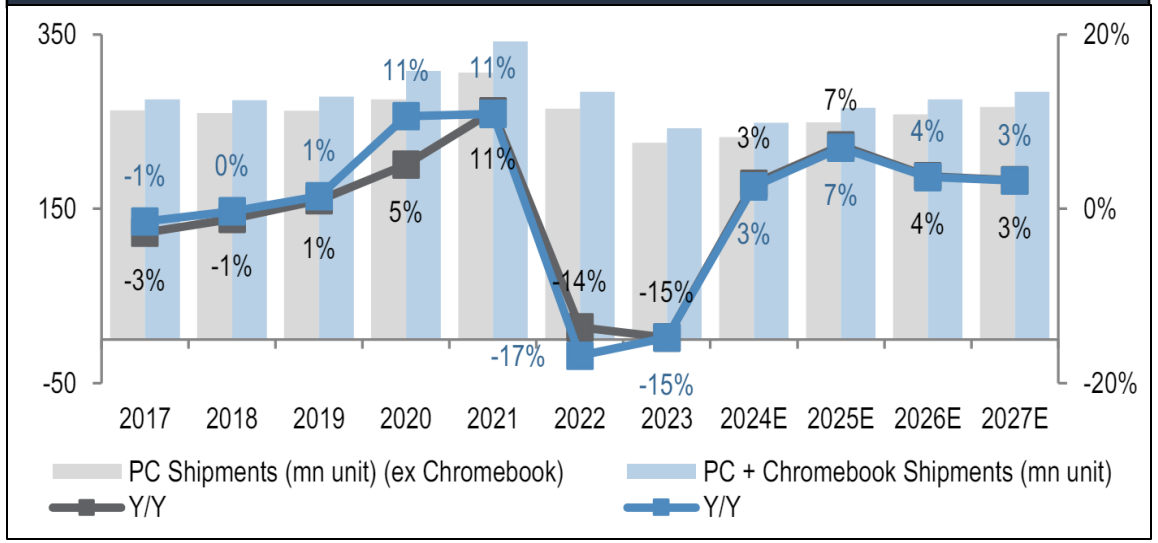
- ✓ 2382 廣達
- ✓ 6669 緯穎
- ✓ 2376 技嘉
- ✓ 2412 中華電
- ✓ 3045 台灣大
- ✓ 2345 智邦
- ✓ 4904 遠傳
- ✓ 5388 中磊

Source: KGI

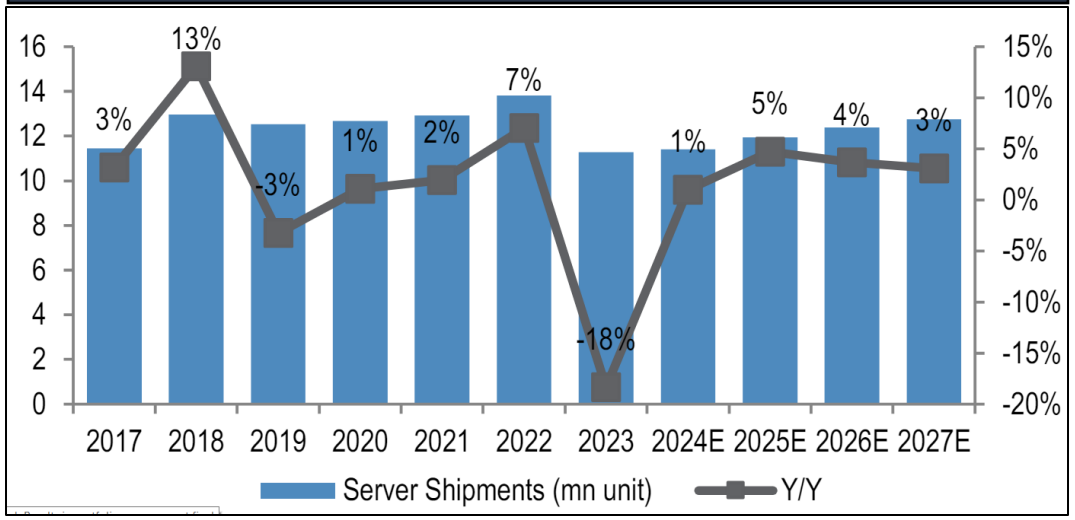
PCs and Servers' AI revenue upticks with slower-than-expected traditional IT recovery

- **Regular servers: Mixed 1H24, but the worst is over.** Intel/AMD/ASPEED's commentary suggested a slowdown in server component demand in 1Q24, which implies lackluster downstream server shipments, near term. We believe the macro uncertainty has pushed out server procurement and expect a seasonal rebound in 2H24. The key overhang is the potential AI server cannibalization impact. Overall, we estimate 1% global server shipment growth this year with the high-end AI server shipment mix up to 5% vs. 2% last year.
- **PCs: Wait for AI PC launch in 2H24/2025.** PC brands/ODMs have suggested weak end demand in 1H, while PC IC vendors have seen rush orders in 1Q. We attribute the divergence to a strategic inventory build up given our expectations of continued component price hikes in coming quarters. There could be downside to upstream vendors if end demand remains muted in 2Q24. Despite a disappointing 1H24, we expect an improving YoY PC trend from 2H24 onwards helped by AI PC launches and Win-10-driven replacement demand.

PC shipments and YOY trend



Server annual shipments and YOY trend



Stocks list:

- ✓ 2317 鴻海
- ✓ 2382 廣達
- ✓ 6669 緯穎
- ✓ 3231 緯創
- ✓ 2357 華碩
- ✓ 4938 和碩

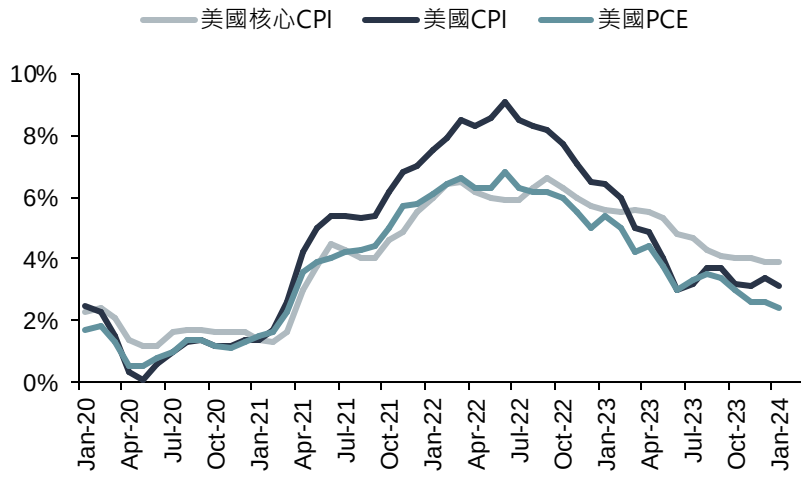
Source: Gartner, J.P. Morgan

Summary of the Strategy Analysis Results

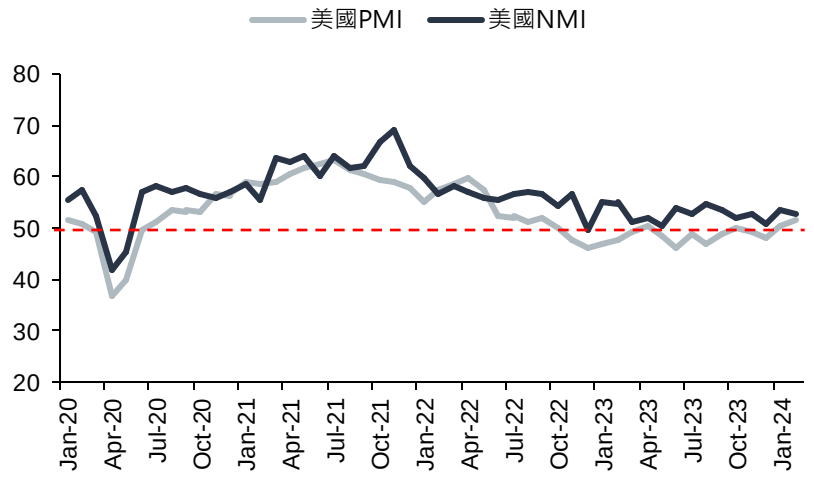
	EW	0053	TWII	VW	MSRPnss	GMVPnss
Final Cumulative Return	**1.0987**	0.40101	0.1488	0.4594	0.6024	0.4707
Annualized Return	**0.2894**	0.1226	0.0487	0.1384	0.1755	0.1414
Annualized Volatility	0.2211	0.2078	0.1831	0.2384	0.1677	**0.1313**
Sharpe Ratio	**1.3087**	0.5899	0.2659	0.5805	1.0462	1.0773
Max Drawdown	-0.1899	-0.3046	-0.2892	-0.2936	-0.2079	**-0.1587**
5% of VaR	-0.0676	-0.0949	-0.0792	-0.0960	-0.0561	**-0.0536**
Appraisal ratio (TWII / 0053)	**0.5792 / 0.404**			0.3637 / 0.0269	0.3741 / 0.2528	0.3311 / 0.2528

美國降息將利於整體股市環境，且總經數據顯示全球外需有復甦跡象，預期以出口為主的台灣半導體產業受惠最大

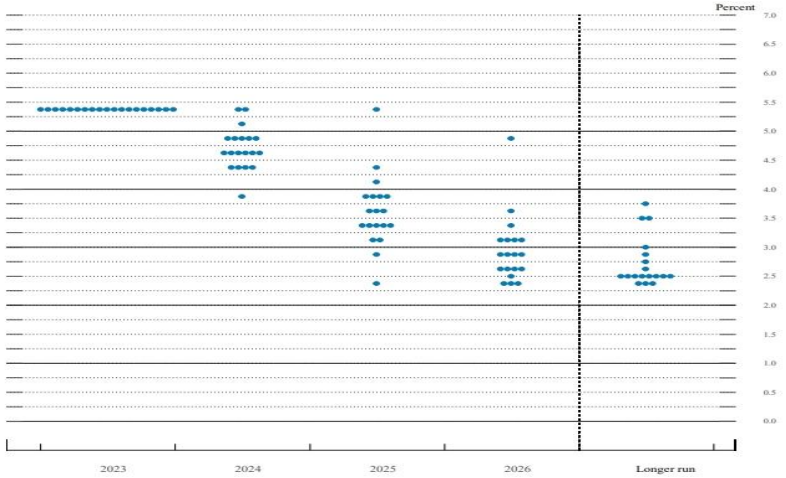
美國通膨已有降溫跡象



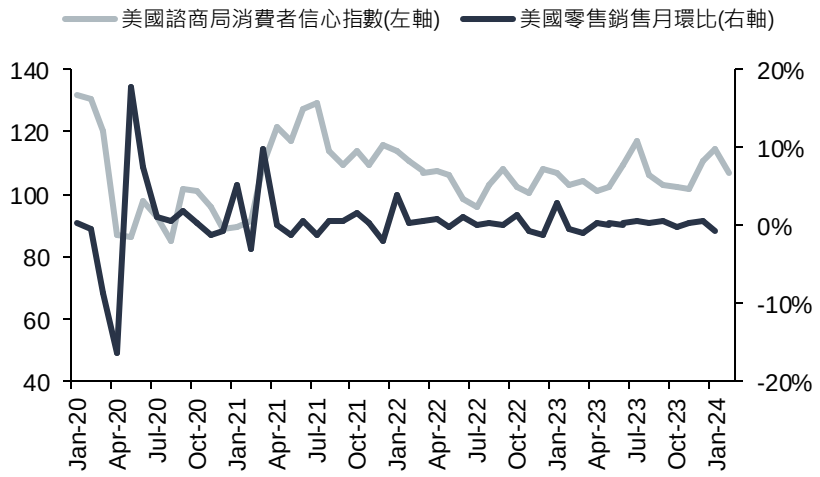
全球外需及美國內需景氣已有復甦跡象



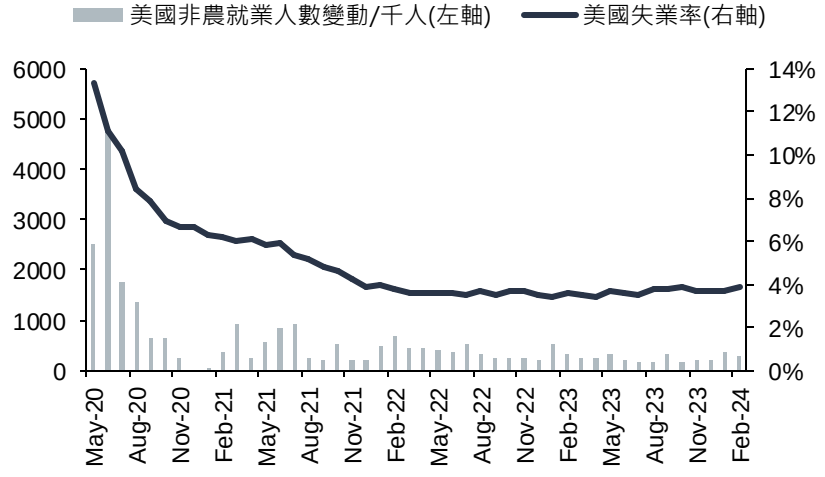
點陣圖預測2024年將降息2-3碼



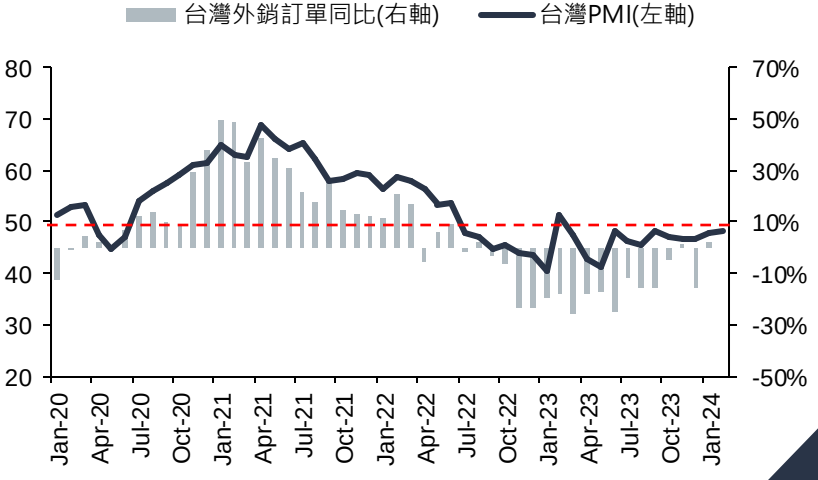
美國消費狀況較穩定



美國勞動狀況穩健未有景氣衰退跡象



台灣製造業及出口狀況仍未完全復甦



資料來源：FOMC，國發會、經濟部統計處、Bloomberg

政府積極推廣綠色金融方案，並改善 ESG 基金資訊揭露及監理原則，促進永續投資的發展

推出綠色金融行動方案

台灣近年陸續推出綠色金融行動方案，2017 年的 1.0 版本著重在鼓勵金融業對「綠能」產業的投資及融資、建置綠色債券市場、鼓勵投信發行 ESG 主題型共同基金及 ETF、推出台灣永續期貨等，2020 年的 2.0 版本則將投資產業擴大至「綠色及永續發展」產業，並繼續推行 1.0 內容，2022 年再推出 3.0 版本，並將重點更擴大至金融業揭露碳排資訊及減碳目標、建置ESG及氣候相關資料庫、永續金融證照、永續金融評鑑等等，等同於將影響力從金融業擴散出去，讓所有企業更注重永續議題，亦使台灣近年以ESG 為主題的基金數量快速增加，統計至目前為止已達48檔。

頒布 ESG 基金監理原則

為健全永續投資的發展，避免 ESG 基金有漂綠的情形，近年陸續頒布《有關證券投資信託事業發行 ESG 相關主題證券投資信託基金之資訊揭露事項審查監理原則》與《ESG 相關主題之境外基金資訊揭露事項審查監理原則》，用以規範業者發行 ESG 基金時，須訂定並公開揭露投資目標與衡量標準、策略與方法、投資比例配置、參考績效指標等等資訊。

建置 ESG 基金專區

金管會為了強化 ESG 基金資訊的揭露，讓集中保管結算所在基金資訊觀測站中建置 ESG 基金專區，基金發行者除需揭露主管機關要求事項，亦要揭露發行資訊、公開說明書及定期報告連結或文件，而所有掛名 ESG 及永續相關的基金都須經金管會審查通過才可放入專區中，增加 ESG 主題基金的資訊透明度及可信度，促進永續投資的發展。

資料來源：金管會、台灣集中保管結算所、行政院、基金資訊觀測站

氣候變遷與環境永續不再是紙上談兵，無論是國家、企業或個人發展，都必須兼顧社會及環境正義

極端氣候

2022 年，世界衛生組織歐洲區域辦事處表示，過去幾十年來歐洲已有數十萬人因極端高溫而喪命，如今歷史重演，又伴隨嚴重乾旱與森林大火的發生，導致英、法等國水資源呈現匱乏，以及西班牙、葡萄牙等國居民遭強制遷移，嚴重打擊民生經濟的發展；此外，根據美國國家氣象局數據顯示，2022 年 6 月，熱浪席捲美國中南部多數地區，德州達拉斯出現攝氏 42.7 度高溫，打破了2018年 42.2 度的紀錄，且加利福尼亞中部及美國東北部分地區也都處於高溫警示之下，影響全美將近三分之一的人口，彰顯各國攜手對抗氣候變遷刻不容緩。

永續發展

世界環境與發展委員會於 1987 年發表「我們共同的未來」（Our Common Future）報告書，對於永續發展（Sustainable Development）的定義為：「既能滿足當代的需要，而同時又不損及後代滿足其需要的發展模式。這個概念深刻改變人們對於經濟發展與環境保護之間關係的認識，成為國際社會應對環境挑戰和促進經濟社會發展並行不悖的基礎理念，亦促使各國政府加強國際合作與協調，共同應對氣候變遷、生物多樣性喪失、資源枯竭等議題。

國際倡議

聯合國於 2015 年 9 月通過《變革我們的世界：2030 年可持續發展議程》，該文件列出 17 項目標及 169 項指標作為永續發展目標（Sustainable Development Goals, SDGs），提供未來 15 年全球共同合作、邁向永續的指引方針。永續發展目標橫跨「經濟成長」、「社會進步」與「環境保護」三大面向，致力於消除貧窮，同時滿足經濟成長、教育、衛生、社會保護和就業機會等社會需求，並回應氣候行動與環境保護等議題，藉以實現社會及環境問題間密不可分的願景。

資料來源：世界衛生組織、美國國家氣象局、世界環境與發展委員會、聯合國

企業遵循永續價值的益處

風險管理

對於 ESG 等方面的考量有助於識別和管理潛在的風險。公司在這些方面表現良好通常意味著它們能夠更好地應對未來的挑戰和不確定性

品牌形象和聲譽

透過致力於永續發展，企業能夠建立良好的品牌形象和聲譽，有助於減少潛在的法律和公共關係風險

投資者信任和支持

越來越多的投資者將ESG因素納入其投資決策中。因此，企業能夠贏得投資者的信任和支持，進而吸引更多資本

長期價值創造

永續價值考慮了長期的影響和風險，投資於這樣的公司通常能夠在長期內穩定且持續地創造價值，而非僅尋求短期的利潤

企業社會責任對其 ROA、ROE 有正向顯著的影響

以 2015 年至 2020 年入選台灣就業 99 指數、台灣高薪 100 指數、台灣公 司治理 100 指數之企業做為研究樣本。將同時入選兩個指數或是僅入選一個指數之 成分股與同產業且資產總額相差 100 億以內之非指數成分股公司進行配對。實證結果發現，企業社會責任表現較佳之企業對其 ROA、ROE 有正向顯著的影響

台灣的 ESG 投資組合中，相較於大盤報酬，有較小的風險且表現優異

本文利用 286 家擁有台灣經濟新報 TESG 評比領先之企業，參考既有文獻以有顯著超額報酬的財務信號構成 ESG 股票投資組合，測試這些投資組合是否能產生顯著超額報酬。本文發現，依投資組合績效來看，台灣的 ESG 投資組合中，相較於大盤報酬，有較小的風險且表現優異。因此，ESG 可作為投資股票時的篩選條件之一，若更進一步使用基本面分析進行量化投資，能夠獲得更穩定且更高的投資組合績效。

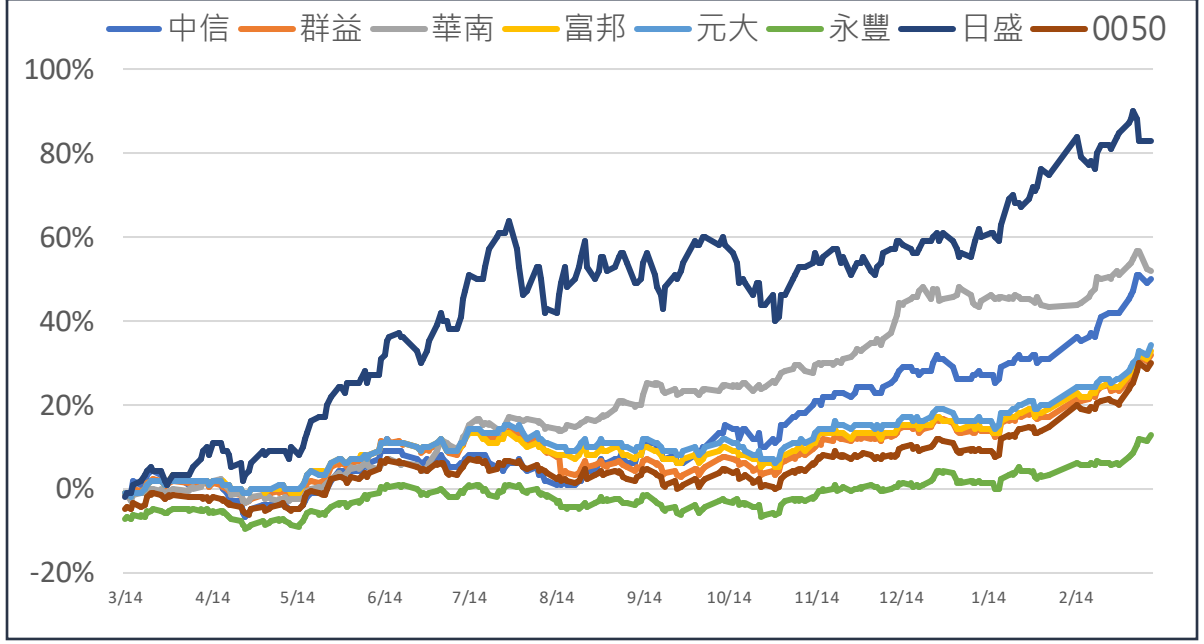
資料來源：林佳靜(2023)，企業社會責任落實度與經營績效之關聯度，管理資訊計算 ，第 12 卷特刊 2 頁 1-10 廖子媛(2023)，運用基本面財務信號於社會責任投資策略之實證，國立臺中科技大學碩士論文

台灣永續主題基金相對於大盤更加強勢

境內ESG股票型基金中不管是在三個月還是一年都表現良好的前10名 (依台幣報酬率為準），累積報酬率優於元大0050。

編號	基金名稱
1	中國信託臺灣ESG永續關鍵半導體ETF基金
2	國泰美國ESG基金
3	日盛台灣永續成長股息基金
4	富達永續全球股票收益基金-N類型累積型
5	群益台灣ESG低碳50ETF基金
6	永豐台灣ESG永續優質ETF基金
7	富邦臺灣公司治理100基金
8	元大臺灣ESG永續ETF基金
9	柏瑞ESG減碳全球股票基金A不配息
10	華南永昌台灣環境永續高股息指數基金B配息

基金累積報酬率走勢圖



	三個月投報率	一年投報率	年化標準差	Sharpe ratio	Beta
最大值	21.57%	80.37%	19.73	1.02	1.18
平均	16.22%	40.18%	13.71	0.69	0.82
最小值	8.72%	19.16%	8.69	0.44	0.45
標準差	4%	18%			

資料來源：MoneyDJ 、TEJPro、Yahoo Finance

台灣永續主題基金成分股產業集中度高

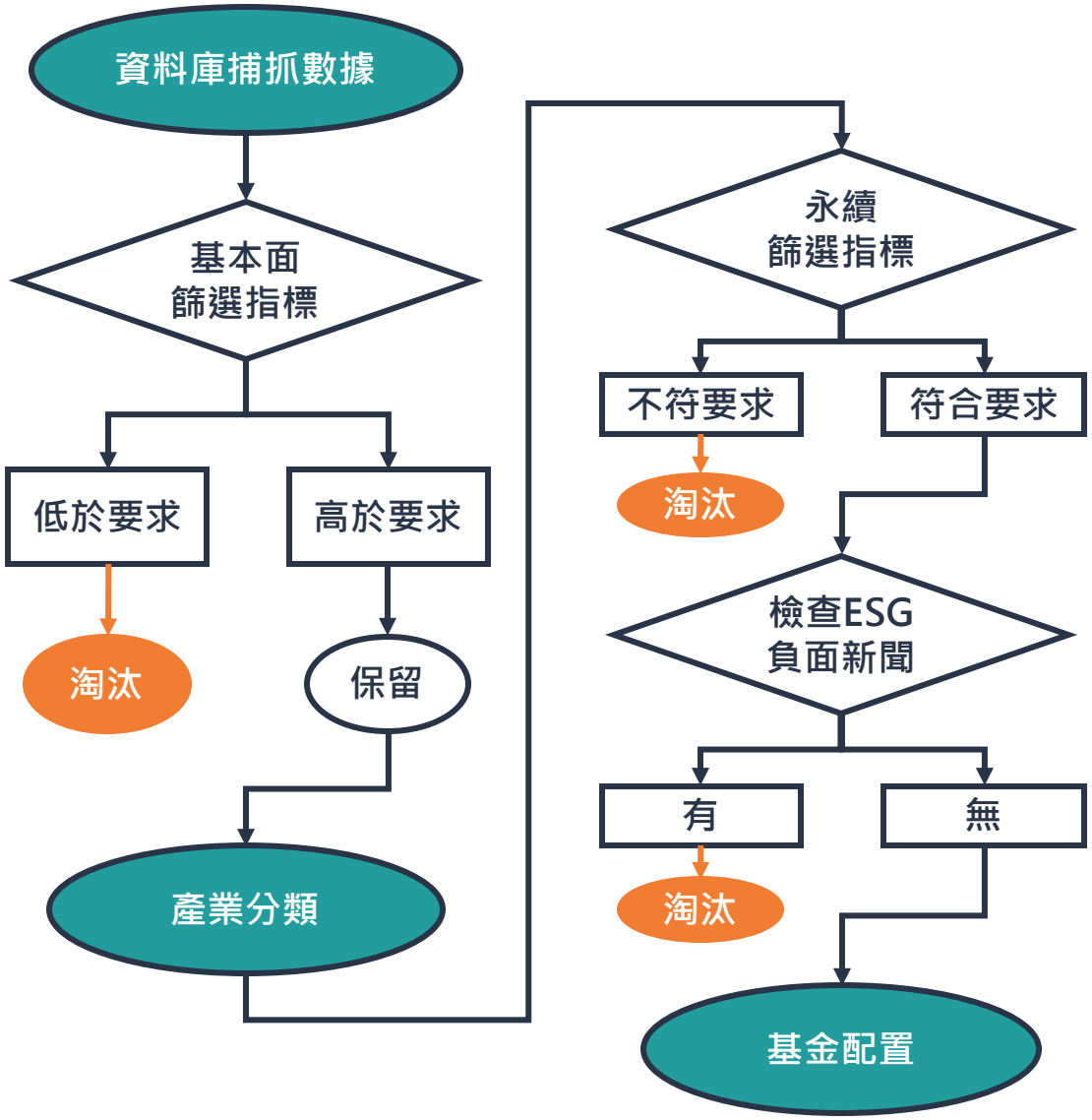
- 根據前面十檔，去除投資國外股票的基金，取每檔成份產業前五名，個股前十名，依據重複率、佔比排序，再依產業區分。經由調查可以發現大部分集中在半導體業、金融保險業、在個股上，重複率也相當高，聚焦在幾檔頗負盛名的公司上。
- 本組發現世芯-KY為多數基金選擇之標的，但是細究其永續層面表現，發現並非永續優質企業，可見部分基金在選擇中，依舊以股票之動能為首要考量。
- 另從十檔基金中，本組亦發現多數永續基金會選擇高股利之股票，本組推論原因為高度在意永續的企業更能長久經營，進一步選擇配發高股利的公司可實現長久的固定收益，間接使得股票型基金產生債券型基金的用處。

重點行業	佔比
半導體業	45-60%
金融保險	10-15%
電腦及週邊設備業	5-9%
通信網路	3-10%
其他電子	3-6%

個股名稱	個股代號	交易所產業名	近一年報酬率 (%)	ROE近三年平均 (%)	每股營收年成長近三年平均 (%)	TESG等級	公司治理評鑑結果
台積電	2330	半導體	52.4344	31.78	17.30	A+	前5%
聯發科	2454	半導體	84.6277	24.54	10.18	A+	前5%
聯電	2303	半導體	6.5145	22.52	7.61	A+	前5%
鴻海	2317	其他電子業	12.5249	9.14	7.44	A	6%至20%
廣達	2382	電腦及週邊	228.5903	18.74	7.54	A+	6%至20%
日月光投控	3711	半導體	55.3777	23.94	17.54	A	6%至20%
聯詠	3034	半導體	50.0567	48.89	11.36	A+	前5%
台達電	2308	電子零組件	9.2405	16.74	12.39	A+	前5%
世芯-KY	3661	半導體	277.7658	22.29	41.44	B	51%至65%
富邦金	2881	金融業	27.5748	11.97	-21.80	A+	6%至20%
國泰金	2882	金融業	10.0695	9.82	-18.00	A	前5%
中信金	2891	金融業	41.7162	10.40	-19.40	A	6%至20%
中華電	2412	通信網路業	7.5415	9.62	2.44	A+	前5%

資料來源：TEJ Pro 、 財報狗

規劃基金配置邏輯，從基本面選擇合適股票，進而篩選永續指標領先公司



資料庫捕抓數據→基本面篩選指標→產業分類

本組藉由財報狗網站，篩選出81檔上市公司股票，選擇上市的原因關注資產的流動性，並參考投資大師巴菲特的選股策略，指定重要基本面指標：ROE、毛利率及營收成長率。

再進一步將81檔上市股票進行產業分類，採用台灣交易所的產業分類及SASB的產業分類，觀察出電子科技產業為大宗，符合台灣資本市場分布情形。

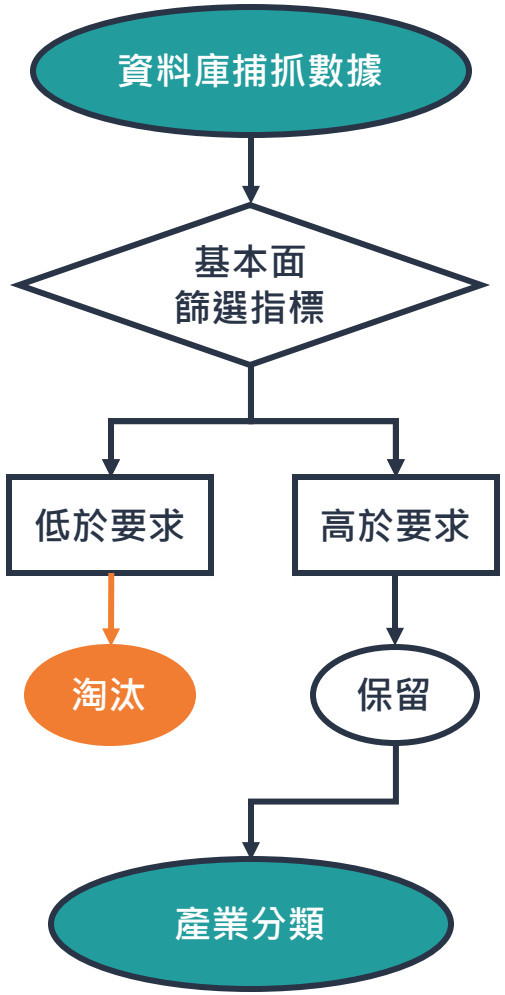
永續篩選指標→檢查負面新聞

本組關注企業永續揭露資訊的行為，認為企業在揭露的過程中亦實現企業的社會責任，並配合台灣公司治理評鑑及TESG永續發展指標，選出19檔符合條件的上市股票，進一步觀察19檔股票是否在近期產生負面ESG新聞，並加以排除，最終選定17檔股票作為本學期基金配置的成分股。

基金配置

最終本組將TESG永續發展指標為A+的9檔上市股票保留為基金成分股，剩餘8檔評比為A則是基金的候補名單。基金的股票權重採用Sharpe ratio作為劃分成分股的比例，建立一檔符合永續指標且無負面ESG資訊的優質股票基金。

採取投資大師巴菲特選股心法，選出81檔上市股票



巴菲特選股法

- 我們根據巴菲特選股法，挑出股票重要基本面指標：ROE、毛利率及營收成長率，我們選擇樣本期間為三年平均資料。
- 關注ROE的原因在於，高ROE公司即表示企業有效運用股東的資本投入。
- 高毛利率則關注企業本身銷貨能力是否可以取得較高的利潤空間
- 營收成長則觀察公司的業務成長動能狀況。

指標	要求
ROE近三年平均 (%)	>15%
毛利率近三年平均 (%)	>10%
每股營收年成長近三年平均 (%)	>10%

篩選結果說明

根據金管會證交所2023年公布最新上市公司數量為997家，根據我們個股篩選機制，藉由財報狗網站查詢功能，最終選定81家上市公司，再根據TEJ資料庫進行彙整，初步篩選發現半導體產業為大宗共12家，其次是電子零組件11家，第三是生技醫療、電腦及周邊與建材營造產業各6家。

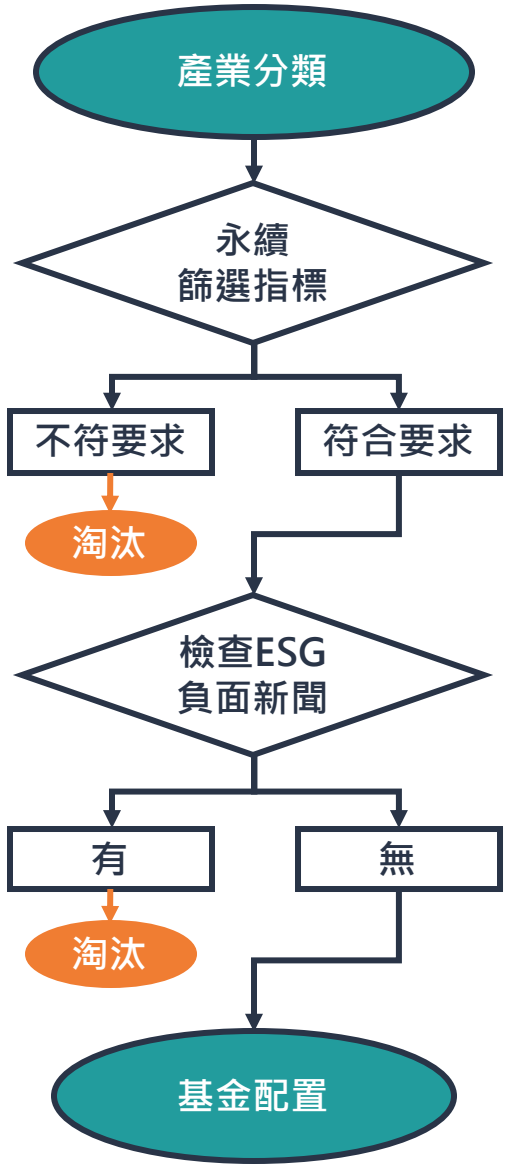
交易所產業別	計數
半導體	12
電子零組件	11
生技醫療	6
電腦及週邊	6
建材營造	6
其他電子業	4
電機機械	4
其他	4
航運業	4
光電業	3
運動休閒	3
通信網路業	3
鋼鐵工業	3
食品工業	2
化學工業	2
綠能環保	2
電器電纜	1
數位雲端	1
紡織纖維	1
塑膠工業	1
電子通路業	1
汽車工業	1
總計	**81**

%	ROE	Gross Margin	Sales Growth
Std	11.69	14.72	51.81
Min	15.12	10.17	10.18
Q1	18.18	21.78	12.62
Med	21.87	29.89	15.75
Q3	27.39	43.12	24.59
Max	76.87	88.85	460.65

SASB主產業	計數
科技與通訊	39
資源轉化	9
運輸	8
公共建設	8
醫療保健	6
消費品	5
提煉與礦產加工	3
食品與飲料	2
金融	1
總計	**81**

資料來源：財報狗、TRJ Pro、

重視企業永續揭露意願，著重台灣在地評比指標



永續指標篩選

- 本組認為辨別企業永續作為有三種方式：1.是否遵循GRI準則；2.是否願意獨立揭露ESG資訊；3.是否有第三方檢驗支持。
- 根據此三點，本組利用TEJ資料庫，辨認企業是否得到GRI認證以及永續報告書是否有第三方查驗。
- 本組亦從第三點延伸，探討企業永續指標狀況，採用國內評比指標：台灣公司治理評鑑，及TESG永續發展指標。

篩選結果說明

- 根據本組設定四個永續指標篩選結果，各有接近半數之企業被排除，交集結果後留下19家台灣上市公司。
- 最終留下半導體業7家，航運業3家，電子零組件2家，其他產業共7家。
- 最後單獨對19家公司進行重大ESG資訊檢驗，近期有2家公司產生ESG負面資訊，在排除後，最終剩下17家台灣上市公司。

指標	說明	篩選標準	結果
是否有GRI認證	GRI提供全球永續報告書的非財務資訊標準，遵循GRI規範可以表達企業是否願意揭露ESG資訊。	根據TEJ資料庫提供「Y」之企業	不符項目有40家
永續報告書是否有查驗	永續報告書是否查驗可以看出企業對自身揭露品質的自信，第三方查驗有助於企業永續資訊的保障。	根據TEJ資料庫提供「Y」之企業	不符項目有39家
台灣公司治理評鑑	公司治理評鑑由金管會2013年建立，希望促成國內企業良性競爭，強化公司治理水平。	需為評選前21%，即B等級以上	不符項目有49家
TESG永續發展指標	由台灣經濟新報（TEJ）自主研發的永續指標，評等台灣上市櫃2402家公司，內容涵蓋ESG三大支柱，共16項議題，超過60項變數建立的量化評估模型。	需為領先群組，即A+與A兩等級	不符項目有54家
ESG事件資料庫	TEJ創立的資料庫，主要提供與ESG相關重大事件資訊，	有負面資訊的拔除	排除2家

資料來源：TEJ Pro

最終選出9檔上市股票作為成分股，8檔候補名單

標的說明

- 本組最終選擇的17檔上市公司股票，依照TESG等級將9檔等級為A+的公司留為本組的基金成分股，等級為A的7檔上市公司股票則成為候補名單。選擇的9檔標的主要產業為半導體業，與市面上的永續基金著重之產業相同，也有台灣本身為半導體產業大國之因素。
- 本組基金與市面上的永續基金不同的地方在於並非追蹤某檔永續指數或是選擇國際知名永續評比機構之評分，而是選用台灣在地指標；所選成分股中，共同的有台積電、聯發科及台達電，主要是半導體產業；另本組並沒有特別追蹤高股息標的。

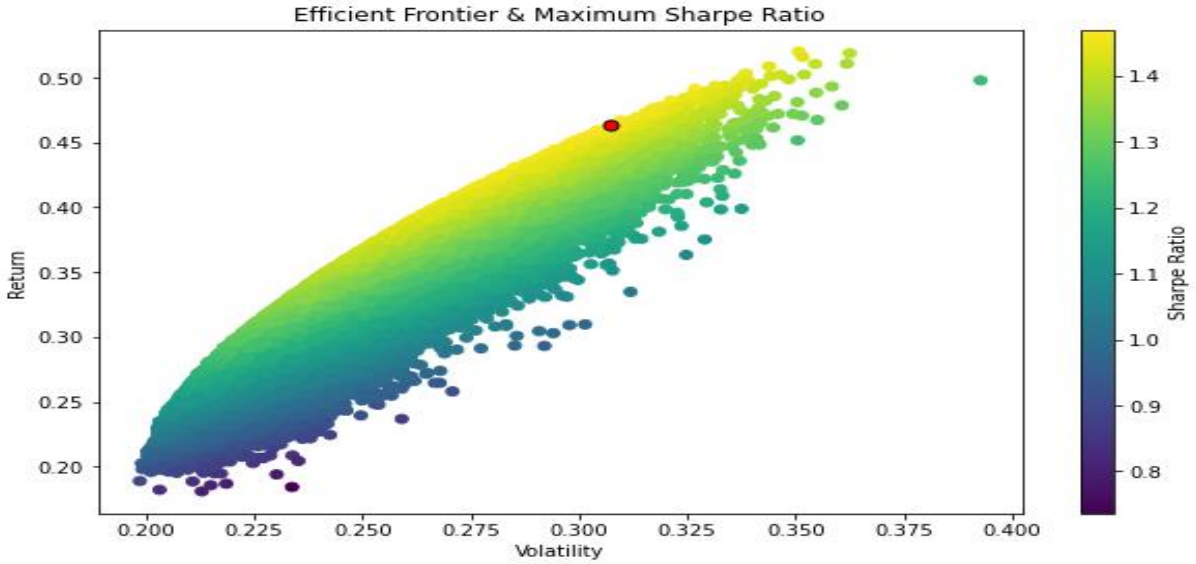
編號	個股名稱	個股代號	交易所產業名	2024/3/8股價	ROE近三年平均 (%)	每股營收年成長近三年平均 (%)	TESG等級	公司治理評鑑結果
1	東和鋼鐵	2006	鋼鐵工業	76.9	16.71	27.28	A+	6%至20%
2	台達電	2308	電子零組件	312	16.74	12.39	A+	前5%
3	台積電	2330	半導體	784	31.78	17.3	A+	前5%
4	智邦	2345	通信網路業	491	38.61	15.56	A+	21%至35%
5	微星	2377	電腦及週邊	188	28.58	14.4	A+	21%至35%
6	聯發科	2454	半導體	1230	24.54	10.18	A+	前5%
7	陽明	2609	航運業	47.55	76.87	23.35	A+	6%至20%
8	創意	3443	半導體	1490	41.73	24.59	A+	前5%
9	新唐	4919	半導體	122	20.44	16.53	A+	21%至35%
10	中碳	1723	化學工業	111.5	18.18	15.75	A	6%至20%
11	長榮	2603	航運業	179.5	73.74	95.61	A	6%至20%
12	萬海	2615	航運業	48.65	62.69	41.05	A	6%至20%
13	智原	3035	半導體	362.5	21.24	29.61	A	6%至20%
14	晶技	3042	電子零組件	103	21.87	16.03	A	6%至20%
15	天鈺	4961	半導體	236.5	27.39	27.16	A	21%至35%
16	旭隼	6409	其他電子業	1620	49.73	11.43	A	6%至20%
17	晶碩	6491	生技醫療	401	23.62	15.27	A	21%至35%

繪製效率前緣，模擬夏普比率最大化之個股權重

權重設立

- 本組利用Python繪製效率前緣落點，設置無風險資產(Rf)為台灣十年期公債殖利率1.22%，股票市場溢酬(ERP)為6.36%，用成分股過去三年股價日報酬率的平均報酬及標準差，隨機生成五十萬次，找出夏普比率最大為1.46。

- 最終我們利用效率前緣，模擬出每單位承擔的風險下，可獲得超額溢酬最大化的股票權重。本組亦有計算股票的市值權重，發現若按照市值權重，台積電權重最大，將得到84.66%；利用夏普比率最大化則是創意為最大權重，將得到21.26%。



編號	個股名稱	個股代號	股價(2024/3/8)	流通在外股數(千股)	市值(千台幣)	SR 最適權重	平均權重	市值權重
1	東和鋼鐵	2006	76.9	730,214	56,153,457	12.30%	11.11%	0.23%
2	台達電	2308	312.0	2,597,543	810,433,416	3.47%	11.11%	3.37%
3	台積電	2330	784.0	25,935,031	20,333,064,304	1.69%	11.11%	84.66%
4	智邦	2345	491.0	560,383	275,148,053	11.79%	11.11%	1.15%
5	微星	2377	188.0	844,856	158,832,928	1.52%	11.11%	0.66%
6	聯發科	2454	1,230.0	1,599,623	1,967,536,290	14.22%	11.11%	8.19%
7	陽明	2609	47.55	3,492,104	166,049,545	17.21%	11.11%	0.69%
8	創意	3443	1,490.0	134,012	199,677,880	21.26%	11.11%	0.83%
9	新唐	4919	122.0	419,765	51,211,330	16.55%	11.11%	0.21%

各標的之間呈現低相關性，基金具有較好的風險分散性

	東和鋼鐵	台達電	台積電	智邦	微星	聯發科	陽明	創意	新唐
東和鋼鐵	1.00								
台達電	0.17	1.00							
台積電	0.16	0.48	1.00						
智邦	0.07	0.38	0.36	1.00					
微星	0.13	0.31	0.34	0.28	1.00				
聯發科	0.16	0.32	0.53	0.35	0.30	1.00			
陽明	0.44	0.15	0.13	0.11	0.13	0.13	1.00		
創意	0.14	0.28	0.35	0.35	0.25	0.33	0.15	1.00	
新唐	0.22	0.20	0.30	0.24	0.21	0.30	0.19	0.40	1.00

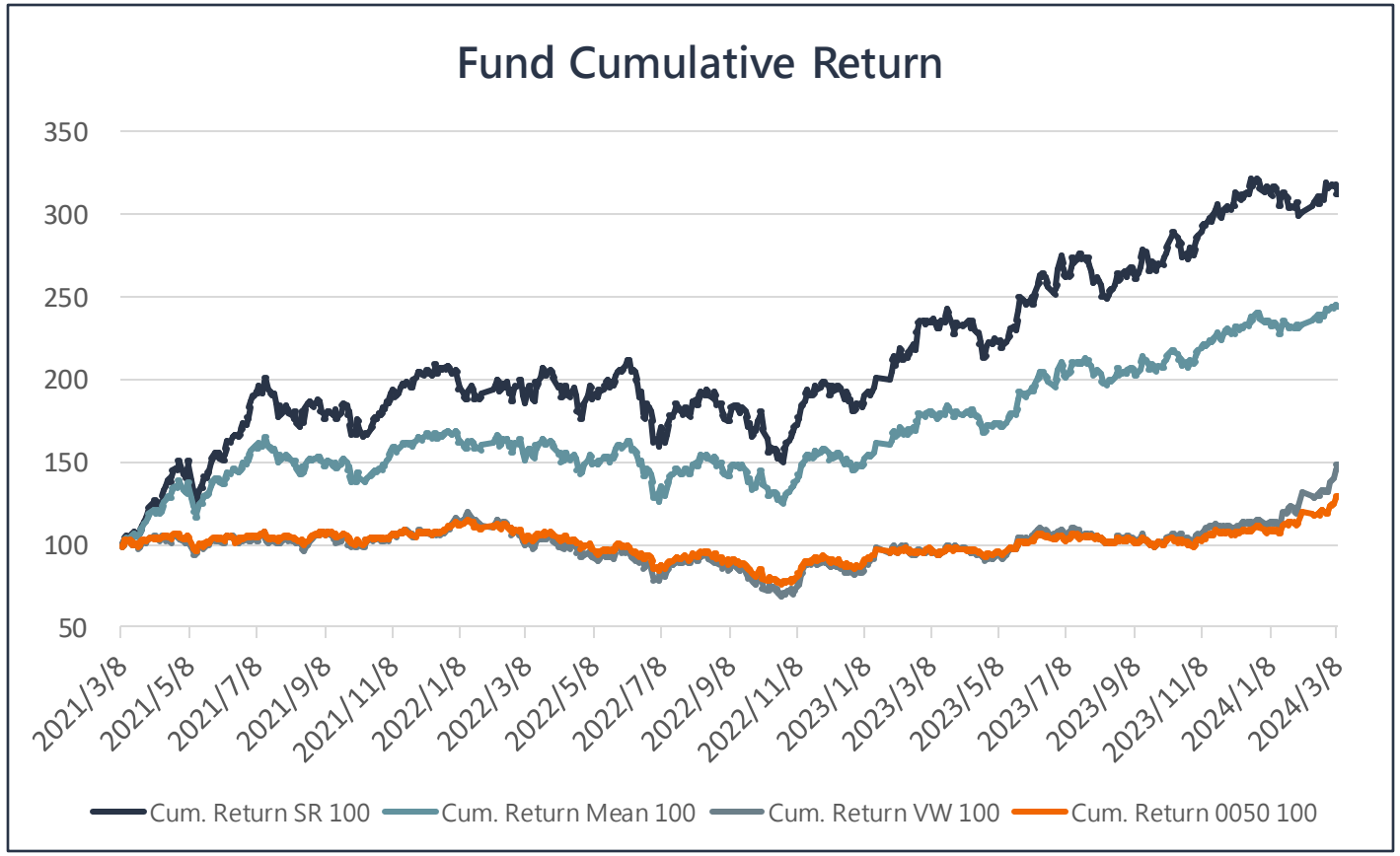
相關性及個股報酬風險

- 從左圖基金內各股票之間的相關性，可以發現彼此呈現低相關性，表示基金具有分散風險的作用，並不會造成各股之間無法牽制的作用
- 下表裡用CAPM計算各股預期報酬率，以及過去三年日報酬率資料得到的波動度，可以發現台積電提供更多的預期報酬，波動性是最低的，但是創意則波動性高，但預期報酬表現相對低。

編號	個股名稱	個股代號	交易所產業別	E(R)	STD
1	東和鋼鐵	2006	鋼鐵工業	6.09%	2.10%
2	台達電	2308	電子零組件	2.50%	1.77%
3	台積電	2330	半導體	7.27%	1.67%
4	智邦	2345	通信網路業	3.80%	2.84%
5	微星	2377	電腦及週邊	5.84%	2.01%
6	聯發科	2454	半導體	6.57%	2.28%
7	陽明	2609	航運業	2.94%	3.61%
8	創意	3443	半導體	2.61%	3.53%
9	新唐	4919	半導體	2.37%	3.34%

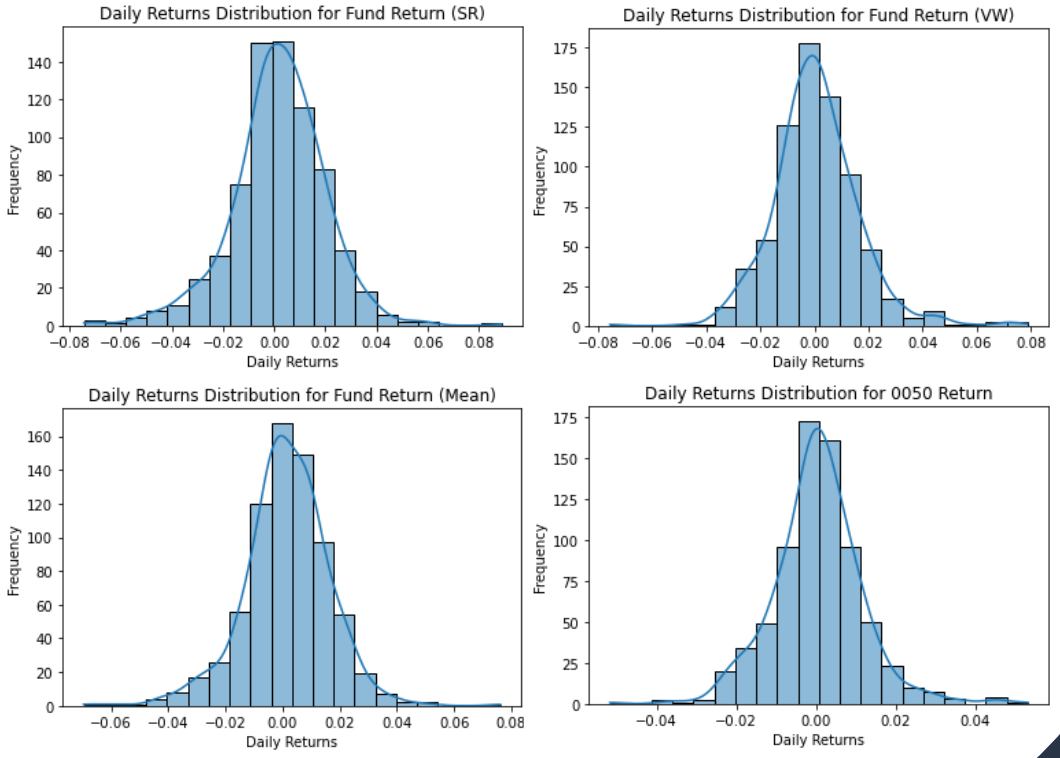
回測歷史三年資訊，遠勝元大0050

	Fund Return (SR)	Fund Return (Mean)	Fund Return (VW)	0050 Return
樣本數n	733	733	733	733
年化報酬率	54.29%	39.83%	18.27%	11.05%
年化標準差	28.96%	23.82%	25.11%	18.00%



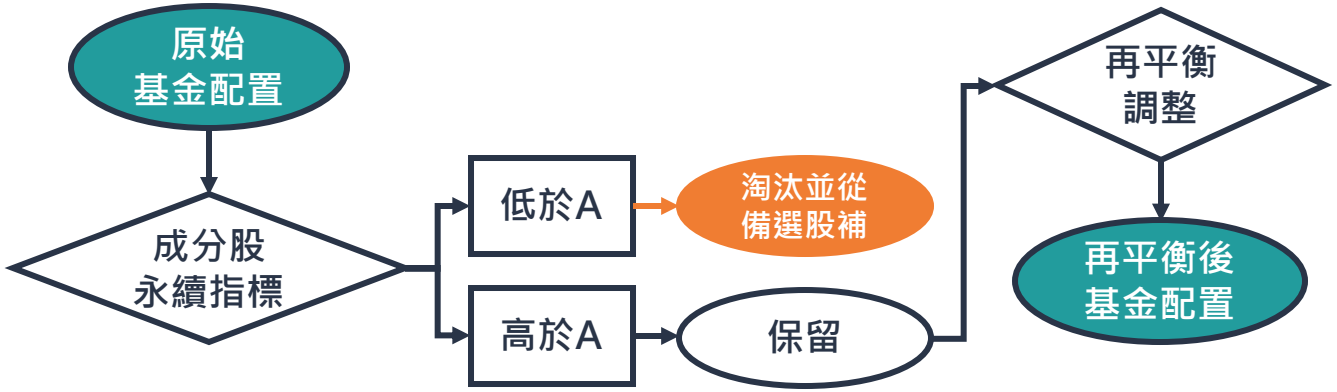
基金報酬與風險

- 本組藉由回測歷史三年數據，亦測試價值加權及平均加權的結果，發現本組選用的夏普比率最大化之權重績效表現最佳，打敗元大0050。
- 從下面的報酬率之相對次數直方圖，可以發現選用夏普比率最高之權重，在報酬率大於0的表現優於其他三者。



重視紀律的管理模式：定期投組績效監控以及投組權重「再平衡」調整

投組管理流程圖(月或季)



監控方法(Monitoring)

- 監控的重要性在於識別趨勢並對投資組合進行調整，以**盡可能**達成最大化報酬與最小化風險(**搭配夏普比率**)。
- 監控方法
  - 注意個股在投組中的占比、價格的變化，定期更新價格資訊總表。(頻率：日)
  - 定期關注個股ESG評分，若A級以下則淘汰並從備選股遞補，重新計算投組夏普比率。(頻率：月或季)

再平衡邏輯(Rebalance Policy)

- 再平衡的重要性：調整投資組合的各資產比重，使投組比重可以與資產配置的策略維持一致。我們使用百分比範圍再平衡法(**Percent-range rebalancing**)，也就是針對投組價值設定百分比臨界值，並以之為依據管理部位比重。
- 再平衡的頻率：以實務上**成本與效益分析**來說，**以月或季為頻率**做績效衡量與投資組合的再平衡最適當。
- 再平衡觸發條件：其中一檔成分股的初始部位占全部的比重浮動超過10%。
  - 例：A股初始投組權重為10%，只要權重大於11%或小於9%則啟動再平衡。
- 再平衡的力度：**視投組中股票價格偏離程度**，我們可以有以下三種在實務上常見的選項
  - 再平衡個股至原先目標權重 (10%)、再平衡到權重範圍邊緣 (11% 或 9%)、再平衡到目標權重與權重範圍邊緣的中間值(10.5% 或 9.5%)。
- 再平衡的替換原則：成分股若有合併、收購、破產、被調查停牌等問題，將會被移除且**由備選成分股中依序遞補**至9檔。

資料來源：2021 CFA PROGRAM LEVEL III REFRESHER READING 13

結論與反思

本組最後利用基本面與永續篩選指標，選出最後的9檔股票，在回測績效的表現上優於元大0050，但是本次報告亦有未完盡的部分：

尚未計算基金的風險水準	並未考慮資產的多元分散性	尚未預測基金未來的報酬走勢
↓	↓	↓
之後利用權重計算基金標準差	可能加入貨幣市場基金	利用蒙地卡羅模擬未來走勢
單一個股投資不得超過10%淨資產	沒有考慮到市場風險的應對	沒有考慮到流動性風險
↓	↓	↓
用貨幣市場基金稀釋權重效果	設計總體衡量指標測試基金反映	觀察個股在市場上的交易量